Natural sparse-parity lower bounds at slowly growing depth

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Abstract

We study the width-first posterior of Gaussian neural networks with hyperbolic-tangent hidden activations, conditioned on a fixed output margin at every Boolean input and on noiseless training labels. For every fixed subcritical width exponent $0 < \gamma < 1$, we prove an $\exp\{ck \log(ed/k)\}$ eventual sample-complexity lower bound for parity on k coordinates when k lies in a fixed polynomial sparsity window and the depth grows as $L(d) = \log_2 \log \log d + O(1)$. The proof gives failure at explicitly specified sample sizes, for the actual posterior and the criterion that the correct label must have probability greater than $2/3$. The new uniform ingredient is a coefficient estimate for iterated arcsine kernels, obtained from an explicit Böttcher coordinate and a uniform contour argument. It permits a depth-independent interpolation budget and a linear parity obstruction on logarithmic active faces. The theorem does not resolve the corresponding natural-scale lower bound at a fixed depth.

1 Model and statement

Let $\mathcal{X} = \{-1, 1\}^d$ carry uniform probability measure. Fix $U \subseteq [d]$ with $|U| = k$ and $f(x) = \chi_U(x) = \prod_{j \in U} x_j$. The learner is not given U. For depth $L = L(d) \geq 2$ and hidden width N, define

$$h^0(x) = (1, x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_N(x) = W^L h^{L-1}(x).$$

There are no additional biases. All weights are independent centered Gaussians, with

$$\operatorname{Var} W^1_{ij} = N^{1-\gamma}/d, \qquad \operatorname{Var} W^s_{ij} = N^{-\gamma} \quad (1 < s < L), \qquad \operatorname{Var} W^L_i = N^{-\gamma-1}. \tag{1}$$

Fix $a > 0$ and $0 < \gamma < 1$. Condition the prior on $|z_N(x)| > a$ for every $x \in \mathcal{X}$ and then on $f(X_i) z_N(X_i) > 0$ for independent uniform training inputs $S = (X_1, \ldots, X_n)$. Write

$$q_{N,S}(x) = \mathbb{P}\{f(x) z_N(x) > 0 \mid |z_N(y)| > a \ \forall y, \ f(X_i) z_N(X_i) > 0 \ \forall i\},$$
$$F_N(S) = \mathbb{P}_x\{q_{N,S}(x) \leq 2/3\}.$$

The width limit is always taken first: d, k, L, a, γ, S are fixed while $N \to \infty$.

Put $D = d + 1$, $\beta = 2^{-(L-1)}$, $p = 2 - \beta$, and $D_r = \sum_{j=0}^{r} \binom{d}{j}$. All logarithms other than $\log_2$ are natural.

Theorem 1.1 (Natural scale at slowly growing depth)**.** *Fix $a > 0$, $0 < \eta_1 < 1 - \eta_2 < 1$, and $0 < b_- \leq b_+ < \infty$. There are constants $\alpha, A, c, C > 0$ and d_0 such that the following holds. For $d \geq d_0$, suppose*

$$d^{\eta_1} \leq k \leq d^{1-\eta_2}, \qquad b_- \leq \beta \log \log d \leq b_+, \qquad \beta = 2^{-(L(d)-1)}.$$

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Choose the largest odd integer $r \le \alpha k$, put $\rho = r/(d+1)$, $H = \log(e/\rho)$, and set

$$n_r = \left\lceil A D_r r H \rho^{-1/400} \right\rceil. \tag{2}$$

Then

$$ck \log(ed/k) \le \log n_r \le Ck \log(ed/k),$$

and for every target support U of size k and every fixed $0 < \gamma < 1$,

$$\mathbb{P}_S \left\{ \liminf_{N\to\infty} F_N(S) \ge \frac{1}{16} \right\} \ge \frac{3}{4}.$$

The conditioning events have positive probability for all sufficiently large widths at every fixed dimension and sample.

The depth condition is equivalent to $L(d) = \log_2 \log \log d + O(1)$ with fixed bounds on the additive term. The result concerns the particular sample sizes (2). It does not assert failure at every smaller sample size, and no monotonicity of posterior risk is used. It implies the stated lower bound for an eventual sample threshold: a guarantee that holds at *every* $n \ge m$ with sample success probability greater than $1/4$ and width-limit confidence failure below $1/16$ must have $m > n_r$. Width is taken first at each d and $L(d)$; no minimum width or interchange of limits is asserted.

1.1 Variational formulation and proof organization

Let $\Psi(t) = 2\arcsin(t)/\pi$ and

$$K(x,y) = \Psi^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d+1}\right).$$

All L^p norms and Walsh coefficients below use the uniform cube probability measure. Let λ_j be the degree-j eigenvalue of the normalized covariance operator $2^{-d}K$. The corresponding energy is

$$\|z\|_K^2 = z^{\mathsf{T}} K^{-1} z = \sum_{V \subseteq [d]} \frac{\widehat{z}(V)^2}{\lambda_{|V|}}.$$

Write $Q_r = K_{>r}$ and $\tau_r = Q_r(x,x) = \sum_{j>r} \binom{d}{j} \lambda_j$. The width-first posterior selects compatible sign functions minimizing $\|z\|_K^2/2$ under

$$|z(x)| \ge a \quad (x \in \mathcal{X}), \qquad f(X_i) z(X_i) \ge a \quad (i \le n). \tag{3}$$

We prove that every minimizer has test error at least $3/8$. The final confidence conclusion is therefore independent of the posterior weights among tied minimizers.

The main additional ingredient beyond a norm bound is transport of the all-input margin. A minimizing field is split into a low-degree head, a middle band, and a sufficiently high-degree tail. Near a sampled nuisance input, the first two parts reduce to a frozen head and a common scalar offset. A recurrence argument and angular concentration force the radial tail profile to return to the head's sign. A center with an incorrect head sign consequently produces a quantitative kernel-energy witness. These witnesses control bounded, normalized wrong-side moments on random active faces. A scalar L^p Bernstein inequality and parity influence then force the restricted head degree to be too large.

2 The kernel rate of the actual network

Let $M = 2^d$, and identify output functions with vectors in $\mathbb{R}^M$. Define

$$\Psi(t) = \frac{2}{\pi}\arcsin t, \qquad K(x,y) = \Psi^{\circ(L-1)}\left(\frac{1+x\cdot y}{d+1}\right). \tag{4}$$

The positive definiteness of K is proved in the next section. For each Boolean function g, put

$$E_K(g) = \min_{g(x)z(x)\geq a\ \forall x} \frac{1}{2} z^T K^{-1} z. \tag{5}$$

The minimum exists since its feasible set is closed and nonempty and the objective is coercive. Replacing the closed inequalities by strict ones gives the same infimum, by scaling a feasible vector by $1+\epsilon$.

Proposition 2.1 (Posterior concentration on minimum-energy signs)**.** *For the actual network* (1)*, at every fixed* $0 < \gamma < 1$*,*

$$\lim_{N\to\infty} N^{-\gamma} \log \mathbb{P}\{g(x)z_N(x) > a\ \forall x\} = -E_K(g). \tag{6}$$

Consequently, after any fixed noiseless training sample, posterior mass tends to one on the compatible functions minimizing E_K*. No assertion about the relative weights of tied minimizers is needed.*

Proof. For hidden layer s, let $Q_N^s(x,y) = N^{-1}\sum_i h_i^s(x)h_i^s(y)$. We first establish, for each fixed $\epsilon > 0$,

$$\mathbb{P}\{\|Q_N^{L-1} - K\| > \epsilon\} \leq C_\epsilon e^{-c_\epsilon N} \tag{7}$$

for all sufficiently large N. Any matrix norm can be used, since M is fixed. Conditional on the preceding layer, the rows of the next hidden layer are independent bounded vectors. Hoeffding's inequality controls all their empirical second moments. Their preactivations have covariance $N^{1-\gamma}Q_N^{s-1}$; at the first layer the covariance is $N^{1-\gamma}(1+x\cdot y)/d$. Because $N^{1-\gamma}\to\infty$, the conditional second-moment map converges to the Gaussian-sign map

$$Q \longmapsto \left[\Psi\left(\frac{Q(x,y)}{\sqrt{Q(x,x)Q(y,y)}}\right)\right]_{x,y}.$$

This follows by dominated convergence for a Gaussian vector, since each scalar variance stays bounded away from zero in a neighborhood of the limiting covariance. The convergence is uniform in such a neighborhood: a compactness subsequence and a continuous square root coupling reduce the assertion to the same dominated-convergence argument. The limiting map is continuous there. Inducting through the fixed number of layers, with smaller fixed tolerances at earlier layers, proves (7). The first-layer limit normalizes the diagonal $(d+1)/d$, giving the kernel in (4).

Conditionally on the hidden layers, the readout is exactly Gaussian:

$$Z_N \mid Q_N^{L-1} \sim \mathcal{N}(0, N^{-\gamma}Q_N^{L-1}). \tag{8}$$

Couple it to $Y_N = N^{-\gamma/2}K^{1/2}\xi$, where $\xi\sim\mathcal{N}(0,I_M)$ is independent of the hidden layers. On a fixed sufficiently small neighborhood of the positive-definite matrix K, the square-root map is Lipschitz. Hence, for $\delta > 0$,

$$\mathbb{P}\{\|Z_N - Y_N\| > \delta\} \leq C_\epsilon e^{-c_\epsilon N} + \mathbb{P}\{\|\xi\| > \delta N^{\gamma/2}/(C\epsilon)\}.$$

First divide logarithms by N^γ and let $N \to \infty$; then let $\epsilon \downarrow 0$. The resulting upper limit is $-\infty$, since $\gamma < 1$. The coupled vectors are therefore exponentially equivalent at speed N^γ.

The fixed-dimensional Gaussian vectors Y_N have large-deviation rate $I(z) = z^T K^{-1} z/2$, directly from their densities. Exponential equivalence transfers the same bounds to Z_N. Applying the upper bound to a closed margin orthant and the lower bound to its interior gives (6), since their infima agree. Finally, there are only finitely many Boolean functions for fixed d. Dividing each compatible orthant probability by their sum proves the posterior concentration claim. □

This is a rate statement for the network itself. It does not require relative-error approximation of rare probabilities, nor convergence in total variation to a Gaussian posterior.

3 Fourier level masses of the kernel

Put $D = d + 1$, $\beta = 2^{-(L-1)}$, and $F = \Psi^{\circ(L-1)}$. The characters $\chi_U(x) = \prod_{j \in U} x_j$ diagonalize the normalized covariance operator $M^{-1}K$, $M = 2^d$. Let λ_j be its eigenvalue at degree j. We first work on the full D-cube with kernel $F(u \cdot v/D)$, whose degree-j eigenvalue is denoted by a_j. Constants in this section may depend on L; Section 4 gives the depth-uniform replacements used in the proof of the main theorem. Notation $b \asymp c$ means two inequalities by positive constant multiples, uniformly in the stated ranges.

Lemma 3.1 (Power coefficients and paired eigenvalues)**.** *Write $F(z) = \sum_{m \text{ odd}} c_m z^m$. Every odd coefficient is positive, $\sum_m c_m = 1$, and*

$$c_m \asymp \frac{\Gamma(m-\beta)}{m!} \asymp m^{-1-\beta} \qquad (m \geq 1 \text{ odd}).$$

The full-cube eigenvalues vanish at even degrees and strictly decrease through odd degrees. They are positive at every odd degree. On the original d-cube,

$$\lambda_{2k} = \lambda_{2k+1} = a_{2k+1} > 0,$$

with a final unpaired eigenvalue if d is even. In particular K is positive definite.

Proof. The series of Ψ has positive coefficients at every odd exponent, with

$$[z^{2k+1}]\Psi(z) = \frac{2}{\pi} \frac{\binom{2k}{k}}{4^k(2k+1)}.$$

Composition preserves strict positivity and oddness. Near $z = 1$ in a slit neighborhood,

$$\Psi(z) = 1 - b(1-z)^{1/2} + O((1-z)^{3/2}), \qquad b = \frac{2\sqrt{2}}{\pi}.$$

Induction gives $F(z) = 1 - A(1-z)^\beta + O((1-z)^{3\beta})$, with $A > 0$; at each composition its leading constant changes from A_t to $b\sqrt{A_t}$, starting from $A_1 = b$. These expansions admit algebraic coefficient transfer [1, Chapter VI]. Here are the analytic-domain details. On the unit circle outside $\{-1, 1\}$, positivity of every odd coefficient gives $|\Psi(z)| < 1$: equality in the triangle inequality would require $z^2 = 1$. Hence the iterated functions extend analytically through every compact arc away from these endpoints. At 1, restrict $\arg(1-z)$ to a closed proper subinterval of $(-\pi, \pi)$. Each square-root composition reduces this angular range, and its positive leading coefficient keeps its image in the slit neighborhood for the next arcsine composition. Oddness gives the identical

argument at -1. These local sectors and the compact arcs provide the required two-singularity indented domain. The contributions of 1 and -1 add at odd exponents, giving

$$c_m \sim \frac{2A}{-\Gamma(-\beta)} m^{-1-\beta} \quad (m \to \infty \text{ odd}).$$

Together with positivity of the finitely many initial coefficients and the gamma-ratio asymptotic, this proves the coefficient comparisons. For $L = 2$ the explicit formula and Stirling's bounds suffice instead.

Let $p_m(j)$ be the probability that the simple coordinate-flip walk on the D-cube reaches a specified distance-j vertex in m steps. Expanding $(u \cdot v/D)^m$ shows that its degree-j Fourier coefficient is $p_m(j)$. Thus $a_j = \sum_{m \text{ odd}} c_m p_m(j)$. For the walk recursion,

$$\begin{aligned} D[p_{m+1}(j) - p_{m+1}(j+2)] = j[p_m(j-1) - p_m(j+1)] \\ + (D-j-2)[p_m(j+1) - p_m(j+3)]. \end{aligned}$$

Induction from time zero gives nonincreasing probabilities through each parity class; terms with zero boundary coefficients are omitted. The term $m = j$ makes the odd-eigenvalue inequalities strict and proves positivity. Even-degree eigenvalues vanish by parity. Restricting the distinguished coordinate to $+1$ gives $\lambda_q = a_q + a_{q+1}$, proving the claims. □

For odd r, the exact combined multiplicity of reduced degrees $j-1, j$ is $\binom{d}{j-1} + \binom{d}{j} = \binom{D}{j}$. Consequently

$$D_r = \sum_{\substack{j \le r \\ j \text{ odd}}} \binom{D}{j}, \quad \tau_r = \sum_{\substack{j > r \\ j \text{ odd}}} \binom{D}{j} a_j, \quad \lambda_{r+1} = a_{r+2}. \tag{9}$$

The odd cutoff is essential for these exact identities.

Lemma 3.2 (Sharp level and tail estimates)**.** *Uniformly for odd $1 \le j \le D/4$ and odd $1 \le r \le D/8$,*

$$\binom{D}{j} a_j \asymp j^{-1-\beta}, \tag{10}$$

$$\tau_r \asymp r^{-\beta}, \tag{11}$$

$$\frac{\lambda_{r+1}}{\tau_r} \asymp \frac{1}{r\binom{D}{r+2}}, \tag{12}$$

$$s_{2,r} := \sum_{q > r} \binom{d}{q} \lambda_q^2 \le \frac{C r^{-2\beta-2}}{\binom{D}{r+2}}. \tag{13}$$

The same level-mass estimate holds up to $j \le \rho D$ for any fixed $\rho < 1/2$, with constants depending on ρ.

Proof. The gamma integral and Lemma 3.1 give, for odd j,

$$a_j \asymp D^{-\beta} \int_0^\infty u^{-1-\beta} p(u)^j (1-p(u))^{D-j}\, du, \quad p(u) = \frac{1-e^{-2u}}{2}, \tag{14}$$

$$\asymp D^{-\beta} \int_0^{1/2} p^j (1-p)^{D-j} w_\beta(p)\, dp, \tag{15}$$

where $w_\beta(p) = [-\frac{1}{2}\log(1-2p)]^{-1-\beta}/(1-2p)$. Indeed the gamma integral Poissonizes the walk time, and independent Poisson coordinate counts give independent coordinate parities with success

probability $p(u)$. Since an odd-distance vertex has zero probability at even walk times, no extra parity factor occurs. The weight w_β is integrable near $1/2$ and comparable to $p^{-1-\beta}$ on $[0,b]$ for every fixed $b<1/2$.

Fix $\rho<b<1/2$ and take $j\le \rho D$. The integral on $[0,b]$ is comparable to the complete beta integral $B(j-\beta, D-j+1)$. For the lower comparison, a beta random variable with these parameters has mean at most ρ, so Markov's inequality puts at least a fraction $1-\rho/b$ of its mass in $[0,b]$. The identity

$$\binom{D}{j} B(j-\beta, D-j+1) = \frac{\Gamma(D+1)\Gamma(j-\beta)}{\Gamma(j+1)\Gamma(D+1-\beta)} \asymp D^{\beta} j^{-1-\beta}$$

gives the desired mass. On $[b,1/2]$, the binomial factor decreases with p and, after multiplication by $\binom{D}{j}$, is at most $\exp[-D\,\mathrm{KL}(\rho\|b)]$. The remaining integrable weight makes this contribution exponentially small. This proves (10) and its stated extended range.

Sum (14) over the odd distances above r. Dropping the parity restriction and using the binomial Markov bound gives

$$\tau_r \le C D^{-\beta} \int_0^\infty u^{-1-\beta} \min\{1, Du/(r+1)\}\, du \le C(r+1)^{-\beta}.$$

For the lower bound, sum (10) over the odd j in $(r, 2r+2]$, using a fixed cutoff strictly below $D/2$ as needed at the endpoint. This yields $cr^{-\beta}$; bounded r are absorbed by a change of constants. Equation (12) follows from (10), (11), and (9).

For the second moment, use paired multiplicities to write $s_{2,r} = \sum_{j>r,\ j \text{ odd}} \binom{D}{j} a_j^2$. Up to an odd cutoff b_D near $D/4$, its terms are at most $Cj^{-2\beta-2}/\binom{D}{j}$. Consecutive odd inverse binomial coefficients have ratio

$$\frac{(j+1)(j+2)}{(D-j)(D-j-1)},$$

uniformly less than a fixed constant below one in this range. The sum is therefore bounded by a constant times its first term, at $j=r+2$. Beyond b_D, monotonicity gives the bound $a_{b_D}\tau_{b_D} \le CD^{-2\beta-1}/\binom{D}{b_D}$, which is exponentially smaller than the claimed bound uniformly for $r \le D/8$. This proves (13). □

Finally, two-step binomial ratios give, uniformly for odd $r \le D/8$,

$$\binom{D}{r} \le D_r \le C\binom{D}{r}, \qquad \frac{D_r}{\binom{D}{r+2}} \le C(r/D)^2. \tag{16}$$

All constants used here can be fixed independently of a subsequently chosen small positive limit $r/d \to \alpha$.

Corollary 3.3 (Lower bounds used by the interpolation)**.** *For sufficiently large d,*

$$\lambda_0 \ge c/d, \qquad \lambda_j \ge \frac{cd^{-1-\beta}}{\binom{d}{j}} \quad (0 \le j \le .49d), \qquad v := \lambda_{\min}(K) \ge c(d\log d)^{-\beta}.$$

Also $v \to 0$.

Proof. The first bound follows from the level estimate at full-cube degree one. For odd reduced degree j, use $\lambda_j = a_j$ and $\binom{D}{j}/\binom{d}{j} = D/(D-j)$. For even j, use $\lambda_j = a_{j+1}$ and $\binom{D}{j+1}/\binom{d}{j} = D/(j+1)$. The extended level bound with any fixed cutoff between .49 and $1/2$ then gives the stated uniform lower bound, including $j=0$.

For any odd full-cube degree $j \le D$, restrict the integral (14) to $u \ge \frac{1}{2}\log(2D)$. In this range $p(u) \ge (1 - 1/(2D))/2$ and $1 - p(u) \ge 1/2$, so $p(u)^j(1-p(u))^{D-j} \ge c2^{-D}$. Integrating $u^{-1-\beta}$ gives $a_j \ge c2^{-D}D^{-\beta}(\log D)^{-\beta}$. Every reduced eigenvalue equals an odd a_j, and the matrix eigenvalues are M times the normalized ones. Thus $v \ge c(d\log d)^{-\beta}$. Finally, for any fixed small positive cutoff fraction, the minimum normalized eigenvalue is at most the average in its tail, whence $v \le M\tau_r/(M - D_r) = O(d^{-\beta})$. This proves $v \to 0$. □

4 Kernel estimates uniform in depth

The preceding fixed-depth estimates alone do not justify increasing L with d. We now prove the required uniform replacements. Put $N_* = L - 1$, $\beta = 2^{-N_*}$, and $a_* = \pi^2/8$; this a_* is unrelated to the output margin a.

Lemma 4.1 (Uniform endpoint coefficient transfer)**.** *There are absolute $C_0, C_1, c, C > 0$ such that, for every dyadic $0 < \beta \le 1/2$ and every odd $j \ge \exp(C_1/\beta)$,*

$$c\beta j^{-1-\beta} \le [z^j]\Psi^{\circ N_*}(z) \le C\beta j^{-1-\beta}.$$

More precisely, with $\delta_\beta = \exp(-C_0/\beta)$, the coefficient is

$$\frac{2a_*^{\beta-1}}{-\Gamma(-\beta)}\frac{\Gamma(j-\beta)}{\Gamma(j+1)} + O(\beta j^{-1-3\beta}) + O(e^{-cj\delta_\beta}),$$

with absolute constants.

Proof. Let $h(q) = 1 - \cos(\pi q/2) = a_* q^2 + O(q^4)$. On a fixed small disk define

$$\phi(q) = a_* q \exp\left\{\sum_{t\ge 0} 2^{-t-1}\log\frac{h(h^{\circ t}(q))}{a_* h^{\circ t}(q)^2}\right\}.$$

The quotient is analytic, even, and $1 + O(q^2)$. Quadratic contraction makes the series locally uniformly convergent, with its derivatives. Reindexing gives $\phi(h(q)) = \phi(q)^2$. The function ϕ is odd, real analytic, and $\phi'(0) = a_*$. Its inverse ψ is odd analytic on another fixed disk and $\psi'(0) = a_*^{-1}$. The endpoint map $g(q) = 1 - \Psi(1-q)$ is the inverse branch of h. Consequently, on the principal slit sector,

$$1 - \Psi^{\circ N_*}(1-q) = \psi(\phi(q)^\beta). \tag{17}$$

Choose C_0 large enough that $|q| \le \delta_\beta$ puts $\phi(q)^\beta$ in a fixed compact subdisk of the domain of ψ. On the upper cut $z = 1 + t + i0$, $q = -t - i0$ and oddness gives

$$\Im\Psi^{\circ N_*}(1 + t + i0) = a_*^{\beta-1}\sin(\pi\beta)t^\beta + O(\beta t^{3\beta}), \qquad 0 < t \le \delta_\beta.$$

The factor β in the error is important. Expand the odd series $\psi(w) = \sum_{j\ \mathrm{odd}} b_j w^j$ and use $|\sin(j\pi\beta)| \le j\pi\beta$; the weighted absolute coefficient sum is uniformly bounded on that compact disk. Replacing $\phi(-t)$ by $-a_* t$ contributes only $O(\beta^2 t^{\beta+2})$.

For completeness, the coefficient contour can be chosen uniformly. Every iterate is analytic on $\Omega = \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$: Ψ maps this domain into $-1 < \Re z < 1$ and hence avoids the cuts. There is an absolute C_H such that

$$\sup_{z\in\Omega,\ |z|\le 1+\delta} |\Psi(z)| \le 1 + C_H\sqrt{\delta} \qquad (0 < \delta \le 1).$$

Indeed project to the unit disk, where positivity of the power series bounds the modulus by one, and use the uniform one-half Hölder estimate near the only boundary singularities ± 1. Iterating this inequality from $\delta = \delta_\beta$ bounds all iterates on $|z| = 1 + \delta_\beta$ by two after increasing C_0. Deform Cauchy's coefficient contour to this circle and the two cut segments. The outer contribution is $O(e^{-cj\delta_\beta})$; oddness makes the cut contributions add at odd j. They equal

$$\frac{2}{\pi} a_*^{\beta-1} \sin(\pi\beta) \int_0^{\delta_\beta} t^\beta (1+t)^{-j-1}\, dt + O\left(\beta \int_0^{\delta_\beta} t^{3\beta}(1+t)^{-j-1}\, dt\right).$$

Extend the leading integral to infinity. The beta integral and Euler's reflection formula give the displayed expansion. Its relative error is small for $j \geq \exp(C_1/\beta)$ with C_1 sufficiently large. □

Lemma 4.2 (Uniform spectral and cusp bounds)**.** *For odd $r \geq \exp(C_1/\beta)$, $r \leq D/8$, the bounds in Lemma 3.2 hold with absolute constants after inserting a factor β in the level mass and eigenvalue-to-tail ratio and a factor β^2 in the second-moment upper bound. In particular,*

$$\tau_r \asymp r^{-\beta}, \qquad \lambda_{r+1}/\tau_r \asymp \frac{\beta}{r\binom{D}{r+2}}.$$

For the matrix eigenvalue $v = \lambda_{\min}(K)$,

$$v \geq c(d \log d)^{-\beta}, \qquad \lambda_0 \geq \beta^{\log_2(\pi/2)}/D.$$

Uniformly at every distance and every spectral cutoff $t \geq 1$,

$$1 - K(x,y) \leq C(\operatorname{dist}(x,y)/d)^\beta, \qquad \tau_t \leq C(t+1)^{-\beta}.$$

Proof. In the positive walk expansion of the degree-j eigenvalue, all walk times below j vanish. Lemma 4.1 and the gamma integral therefore give, uniformly for $j \geq r$ odd,

$$a_j \asymp \beta D^{-\beta} \int_0^\infty u^{-1-\beta} p(u)^j (1-p(u))^{D-j}\, du, \qquad p(u) = (1-e^{-2u})/2.$$

The beta-integral proof of the level estimate is uniform for $0 < \beta \leq 1/2$. The integral away from $p = 0$ has at most a factor $1/\beta$, canceled by the prefactor, and its binomial term is exponentially small. The resulting level mass is $\binom{D}{j} a_j \asymp \beta j^{-1-\beta}$ up to any fixed fraction below $D/2$. For the tail upper bound, integrate $\beta u^{-1-\beta} \min(1, Du/r)$; its constant is uniform. For the lower bound integrate over the whole interval $u \geq Cr/D$: a $\mathrm{Bin}(D, p(u))$ variable exceeds r with odd parity with probability bounded below. This gives $cr^{-\beta}$; summing only the first dyadic shell would incorrectly lose a factor β. The paired ratio and second moment follow from the same binomial ratios as in Lemma 3.2. The exponentially small far-degree remainder is absorbed uniformly: the threshold $r \geq \exp(C_1/\beta)$ implies $\beta^{-1} \leq C \log r$, whereas the remaining binomial gap is exponential in d.

To bound v, use paired monotonicity and an odd full-cube degree near D. Restrict the integral to $u \geq \frac{1}{2}\log(2D)$, where its binomial monomial is at least $c2^{-D}$. The integral of $\beta u^{-1-\beta}$ cancels the factor $1/\beta$. Multiplication by 2^d gives the matrix bound. The constant mode is bounded below by the linear Taylor term $(2/\pi)^{N_*}/D$.

For the cusp, (17) gives the claimed uniform bound when $1 - (1 + x \cdot y)/D \leq \delta_\beta$. Outside that range use the bound two and $\delta_\beta^\beta = e^{-C_0}$. The full tail above $\exp(C_1/\beta)$ was just proved; below that threshold the trivial bound one is at most $e^{C_1}(t+1)^{-\beta}$. □

All subsequent uses of depth-dependent fixed-depth estimates can now be made uniform. The first-shell Gram proof uses only the upper shell mass, lower tail, paired ratios, and positive monotonicity. Radial second moments use denominators $2-\beta$, bounded away from zero. The angular-envelope Cauchy weights below use the fixed exponent $1/4$, so no factor $1/\beta$ occurs in their sum. Representation, Gaussian concentration, VC, and recurrence constants are independent of depth. Every displayed positive-power error exponent is smallest at $\beta = 1/2$. The fine covariance witnesses use only uniform *upper* bounds, so their lower energy constants remain uniform. The polynomial sparsity window absorbs all explicit logarithmic factors. These observations will be used in the final assembly.

5 A uniform globally separated comparator

Lemma 5.1 (Exact interpolation by point masses)**.** *For any consistent training labels and any $a > 0$, there is a globally margin-separated fitting field with*

$$\|z_{\mathrm{comp}}\|_K^2 \le Ca^2 \left[d\beta^{-\log_2(\pi/2)} + n(d\log d)^\beta \right].$$

If $\beta \log\log d \le B_0$ and $n \gg d^2$, this is at most $C_{B_0} n a^2 d^\beta$. The statement is deterministic.

Proof. Start from the constant field $-a$ and add $2a\delta_x$ at each distinct positively labeled training input. The resulting field is exactly $\pm a$ on the whole cube and fits all labels. The constant has energy a^2/λ_0. If v is the minimum matrix eigenvalue, the added vector has Euclidean squared norm at most $4a^2 n$, hence RKHS squared norm at most $4a^2 n/v$. Apply Lemma 4.2 and $\|u+v\|_K^2 \le 2\|u\|_K^2 + 2\|v\|_K^2$. Strict margins, when needed, follow by an arbitrarily small common scaling. □

6 One-sided squared sampling with one logarithm

The sampling estimate needed below concerns a squared one-sided loss. It follows from a metric estimate in the proof of Lee's chaining theorem [2, Lemma 2.12, equation (2.20)]. We spell out the comparison: a first-power ReLU sampling theorem alone would not give this result. Throughout this section, C denotes a universal constant that may change from line to line, and $t_- := \max\{-t, 0\}$.

Lemma 6.1 (A squared Lipschitz process)**.** *Let $a_1, \dots, a_m \in \mathbb{R}^D$, and set*

$$R = \max_{i \le m} \|a_i\|_2, \qquad V = \sup_{\|u\|_2 \le 1} \left(\sum_{i=1}^m \langle a_i, u\rangle^2 \right)^{1/2}.$$

For each i, let $\psi_i : \mathbb{R} \to \mathbb{R}$ satisfy $\psi_i(0) = 0$ and $\mathrm{Lip}(\psi_i) \le 1$. For independent uniform signs ϵ_i,

$$\mathbb{E}_\epsilon \sup_{\|u\|_2 \le 1} \left| \sum_{i=1}^m \epsilon_i \psi_i(\langle a_i, u\rangle)^2 \right| \le CR\sqrt{\log(2(m+D))}\, V. \tag{18}$$

Proof. Use the genuine seminorms $N_i(u) = |\langle a_i, u\rangle|$ and the symmetric quasimetric

$$d_A(u,v)^2 = \sum_{i=1}^m N_i(u-v)^2 (N_i(u)^2 + N_i(v)^2).$$

The stronger intermediate estimate established in the proof of [2, Lemma 2.12] is

$$\gamma_2(T, d_A) \le C(\lambda\sqrt{\log(2D)} + \kappa) \sup_{u \in T} \left(\sum_i N_i(u)^2 \right)^{1/2}, \qquad T \subseteq B_2^D, \tag{19}$$

where

$$\lambda = \max_i \big(\mathbb{E} N_i(g)^2\big)^{1/2}, \qquad \kappa = \mathbb{E} \max_i N_i(g), \qquad g \sim \mathcal{N}(0, I_D).$$

Here γ_2 is the generic-chaining functional. This is equation (2.20) of that proof, with the energy normalization (2.17) undone. Seminorms follow from the norm statement by replacing $N_i(u)$ with $(N_i(u)^2 + \varepsilon^2 \|u\|_2^2)^{1/2}$ and then letting $\varepsilon \downarrow 0$; the resulting metrics dominate d_A, and the parameters on the right converge. For our rank-one seminorms, $\lambda = R$ and $\kappa \le R\sqrt{2\log(2m)}$.

Let $X_u = \sum_i \epsilon_i \psi_i(\langle a_i, u\rangle)^2$. Its subgaussian increment metric is bounded by $\sqrt{2}\, d_A$, because

$$\begin{aligned} &\sum_i \Big(\psi_i(\langle a_i, u\rangle)^2 - \psi_i(\langle a_i, v\rangle)^2\Big)^2 \\ &\quad \le \sum_i |\langle a_i, u - v\rangle|^2 \big(|\langle a_i, u\rangle| + |\langle a_i, v\rangle|\big)^2 \le 2 d_A(u, v)^2. \end{aligned}$$

We used both Lipschitz continuity and $|\psi_i(t)| \le |t|$. The subgaussian chaining inequality, together with $X_0 = 0$, bounds $\mathbb{E} \sup X_u$ by $C\gamma_2(B_2^D, d_A)$. Apply the same argument to $-X_u$ and add. Equation (19) now proves (18). □

Remark 6.2 (The normalization in the metric estimate). The proof of [2, Lemma 2.12] passes through a coarse additive constant in its equation (2.26), after normalizing $\sup_T \sum_i N_i^2 = 1$. The homogeneous version (19) follows by retaining the scale in the two geometric terms of that argument. Indeed, in the rank-one case $T \subseteq B_2^D$ implies $N_i(u) \le \lambda$ on T; writing $\|u\|_{\mathcal{N}} = \max_i N_i(u)$ gives $d_A(u, v) \le \sqrt{2}\|u - v\|_{\mathcal{N}}$ on T. Thus $\operatorname{diam}_{d_A}(T) \le C\lambda$ and the volumetric packing constant in its Corollary 2.11 can be taken to be $C\lambda$, since $T \subseteq \lambda B_{\mathcal{N}}$. The additive term is therefore $C\lambda$, absorbed by $C(\lambda\sqrt{\log(2D)} + \kappa)$. The growth-functional portion of the proof is unchanged. If the packing constant is less than one, first rescale the quasimetric to apply that corollary in its stated normalization, and then scale back. Most importantly, throughout that argument the N_i are symmetric seminorms. No assertion that a positive part is a seminorm is used.

Lemma 6.3 (Uniform one-sided squared sampling)**.** *Let (Z, Y) be a random pair with $Z \in \mathbb{R}^D$, $Y \in \{-1, 1\}$, $\mathbb{E} Z Z^{\mathsf{T}} = I_D$, and $\|Z\|_2^2 = D$ almost surely. For independent copies (Z_i, Y_i), let*

$$\Delta_n = \sup_{\|u\|_2 \le 1} \left| \frac{1}{n} \sum_{i=1}^n (Y_i \langle Z_i, u\rangle)_-^2 - \mathbb{E}(Y\langle Z, u\rangle)_-^2 \right|, \qquad h_n = \sqrt{\frac{D \log(2(n + D))}{n}}.$$

Then $\mathbb{E}\Delta_n \le C h_n(1 + C h_n)$. In particular, for any fixed $\varepsilon, \delta > 0$ there is a finite $A_{\varepsilon,\delta}$ such that

$$n \ge A_{\varepsilon,\delta} D \log(2D) \quad \Longrightarrow \quad \mathbb{P}\{\Delta_n > \varepsilon\} \le \delta.$$

The labels may depend arbitrarily on the features.

Proof. Put $Q_n = n^{-1} \sum_i Z_i Z_i^{\mathsf{T}}$. Conditional on the observations, apply Lemma 6.1 to $a_i = Y_i Z_i / \sqrt{n}$ and $\psi_i(t) = t_-$. Here $R = \sqrt{D/n}$ and $V = \sqrt{\|Q_n\|_{\mathrm{op}}}$. Symmetrization gives

$$\mathbb{E}\Delta_n \le C h_n \, \mathbb{E}\sqrt{\|Q_n\|_{\mathrm{op}}}. \tag{20}$$

We also apply the same process lemma with $\psi_i(t) = t$. Since $\mathbb{E} Q_n = I_D$, symmetrization and Jensen's inequality imply, with $A = \mathbb{E}\|Q_n\|_{\mathrm{op}}$,

$$A \le 1 + \mathbb{E}\|Q_n - I_D\|_{\mathrm{op}} \le 1 + C h_n \mathbb{E}\sqrt{\|Q_n\|_{\mathrm{op}}} \le 1 + C h_n \sqrt{A}.$$

Solving this quadratic inequality yields $\sqrt{A} \le 1 + Ch_n$, after increasing C. Substitute into (20). Finally, at $n \ge A_{\varepsilon,\delta} D \log(2D)$ the quantity h_n can be made uniformly small by increasing $A_{\varepsilon,\delta}$. Markov's inequality proves the probability statement. It is a statement for each specified sample size n. □

7 Random tail covariance matrices

Let Q be the covariance kernel retaining only degrees greater than an odd cutoff r, and write $\tau = \tau_r$, $\ell = \lambda_{r+1}$, and $s_2 = s_{2,r}$. This section proves the sample event needed later, without assuming that a Gaussian approximation remains valid after conditioning. The entire argument concerns the exact limiting Gaussian covariance.

Lemma 7.1 (Far-pair decay)**.** *There are constants $c_0, C > 0$ such that, when $r \le D/8$ is odd,*

$$\max_{d/4 \le \mathrm{dist}(x,y) \le 3d/4} |Q(x,y)| \le \rho_r := C\sqrt{D}\, e^{-r/8} + Ce^{-c_0 D}.$$

Proof. The full-cube Krawtchouk polynomial satisfies

$$\sum_{j=0}^{D} \mathsf{K}_j(h;D) z^j = (1+z)^{D-h}(1-z)^h.$$

If $\delta D \le h \le (1-\delta)D$, put $b = \min(h, D-h)$. On $|z| = t$ the generating function is bounded in absolute value by

$$(1+t^2)^b (1+t)^{D-2b} \le (1+t)^D \exp\left[-\frac{2\delta D t}{(1+t)^2}\right].$$

Cauchy's coefficient bound at $t = j/(D-j)$ and the uniform Stirling lower bound for $\binom{D}{j}$ therefore give, for $1 \le j < D$,

$$\frac{|\mathsf{K}_j(h;D)|}{\binom{D}{j}} \le C\sqrt{D} \exp[-2\delta j(1-j/D)]. \tag{21}$$

For reduced-cube points at distance in $[d/4, 3d/4]$, their augmented full-cube distance lies in $[D/8, 7D/8]$ for large d. Because r is odd, the exact tail is $Q(x,y) = \sum_{j>r,\ j \text{ odd}} a_j \mathsf{K}_j(h;D)$. For $r < j \le D - r$, (21) gives a bound $C\sqrt{D}e^{-r/8}$ after summation, since the total spectral mass is one. For $j > D - r$, monotonicity and an odd cutoff b_D near $D/4$ give

$$\sum_{j>D-r} \binom{D}{j} a_j \le a_{b_D} \sum_{k<r} \binom{D}{k} \le CD^{-\beta-1/2} \exp[-D(\mathsf{h}(1/4) - \mathsf{h}(1/8)) + O(1)].$$

Here $\mathsf{h}(t) = -t\log t - (1-t)\log(1-t)$ denotes the binary entropy in nats. The positive entropy gap absorbs the polynomial factor and proves the claim. Adjust constants for bounded d. □

Lemma 7.2 (A tail Gram deviation bound)**.** *Let $X_1, \dots, X_n$ be independent uniform cube points, with $1 \ll n \le M/4$. Let*

$$B_{\mathrm{near}} = \#\{y : \mathrm{dist}(x,y) \notin [d/4, 3d/4]\} \le 2e^{d\mathsf{h}(1/4)}, \qquad q = 2\log M.$$

The count is independent of x. For every $z > 0$,

$$\begin{aligned} &\mathbb{P}\{\|Q_{SS} - \tau I_n\|_{\mathrm{op}} > z\tau\} \\ &\quad \le \frac{C}{z^2}\left[\frac{q\rho_r}{\tau} + \frac{\sqrt{nqs_2}}{\tau} + \frac{n\ell}{\tau} + \frac{n(1+B_{\mathrm{near}})}{M}\right]^2 + \frac{n^2(1+B_{\mathrm{near}})}{M}. \end{aligned} \tag{22}$$

Here the sampled matrix includes repeated training locations.

Proof. Let $J = Q - \tau I_M$, a zero-diagonal Hermitian matrix. Form a deterministic mask J_{far} retaining only pairs at distances in $[d/4, 3d/4]$. Since Q is positive semidefinite with constant diagonal τ, every entry has absolute value at most τ. Thus

$$\begin{aligned} \|J_{\mathrm{far}}\|_{\max} &\le \rho_r, \\ \|J_{\mathrm{far}}\|_{1\to 2}^2 &\le M s_2, \\ \|J_{\mathrm{far}}\|_{\mathrm{op}} &\le M\ell + \tau + \tau B_{\mathrm{near}}. \end{aligned}$$

For the second inequality, Parseval gives squared column norm $Ms_2 - \tau^2$ for J, and entrywise masking only decreases it. For the third, $\|J\| \le M\ell + \tau$, while the removed matrix has maximum absolute row sum at most τB_{near}. The entropy bound on B_{near} is the elementary binomial tail bound on distances to a fixed cube point.

Let R retain each of the M coordinates independently with probability $p = 2n/M$. The zero-diagonal case of Tropp's random principal-submatrix theorem [3, Theorem 1.1], at moment order two and $q = 2\log M$, states

$$(\mathbb{E}\|RJR\|^2)^{1/2} \le C\left[q(\mathbb{E}\|RJR\|_{\max}^2)^{1/2} + \sqrt{pq}(\mathbb{E}\|JR\|_{1\to 2}^2)^{1/2} + p\|J\|\right].$$

Apply it to J_{far} and substitute the preceding bounds:

$$(\mathbb{E}\|RJ_{\mathrm{far}}R\|^2)^{1/2} \le C[q\rho_r + \sqrt{nqs_2} + n\ell + (n/M)\tau(1 + B_{\mathrm{near}})]. \tag{23}$$

The logarithm here is that of the original dimension M, not the sample size.

To pass to a uniform subset of size n, use a uniform permutation of all cube points, independently of $B \sim \mathrm{Bin}(M, 2n/M)$. Its first B elements form the Bernoulli subset. On $B \ge n$, the first n principal submatrix is a compression of the larger one, so its operator norm cannot be larger. The size-n event is independent of B, and $\mathbb{P}(B \ge n) \ge 1/2$ for large n. Thus Markov's inequality applied to (23) gives the same fixed-size probability bound with a constant factor loss.

Conditional on no collisions, an independent size-n sample is a uniform size-n subset, up to an irrelevant ordering. Collisions have probability at most $n(n-1)/(2M)$. The probability of any sampled pair violating the distance condition is at most $n^2 B_{\mathrm{near}}/M$. On their complement, the sampled matrices J and J_{far} agree. Adding these exceptional probabilities proves (22). □

8 Tail covariance estimates at sparse cutoffs

The original principal-submatrix estimate uses the logarithm of the full cube cardinality. We retain that estimate here. A more precise bound on entries near orthogonal input pairs makes it applicable even when the spectral cutoff is a sufficiently large fixed integer.

Lemma 8.1 (Near-orthogonal entries)**.** *There are constants $a_0, \delta_0, c, C > 0$ such that the following holds. Put $D = d + 1$, let r be odd with $r + 2 \le \delta_0 D$, and let h be the Hamming distance between the augmented inputs $(1, x), (1, y)$. If $a = |D - 2h|/D \le a_0$, then*

$$\frac{|Q_r(x,y)|}{\tau_r} \le [C(a + \sqrt{(r+2)/D})]^{r+2} + Cd^{\beta}e^{-cd}, \qquad Q_r = K_{>r}. \tag{24}$$

Proof. For $1 \le j < D$, the Krawtchouk generating function gives

$$\frac{|\mathsf{K}_j(h; D)|}{\binom{D}{j}} \le [e^{3/2}(a + \sqrt{j/D})]^j. \tag{25}$$

Indeed, set $b = \min(h, D-h)$. On the circle $|z| = t$, the generating function has modulus at most

$$(1+t^2)^b(1+t)^{D-2b} \le \exp(Dt^2/2 + Dat).$$

Cauchy's coefficient bound and $\binom{D}{j} \ge (D/j)^j$, with $t = j/[D(a+\sqrt{j/D})]$, prove (25): both Dt^2 and Dat are at most j.

Choose a_0, δ_0 so small that $e^{3/2}(a_0 + \sqrt{\delta_0}) \le e^{-1}$ and $\delta_0 < 1/8$. On $j \le \delta_0 D$, the derivative of $j \log[e^{3/2}(a + \sqrt{j/D})]$ is at most $-1 + 1/2 < 0$. Consequently (25) is bounded by its value at $r+2$ throughout $r < j \le \delta_0 D$. The sum of the corresponding positive spectral masses $\binom{D}{j} a_j$ is at most τ_r. For $\delta_0 D < j \le (1-\delta_0)D$, the central-distance estimate (21) bounds the normalized Krawtchouk polynomial by $C\sqrt{D}e^{-cD}$; its polynomial factor can be absorbed by decreasing c. For $j > (1-\delta_0)D$, choose an odd b_D near $D/4$. Monotonicity of a_j and Lemma 3.2 bound the remaining mass by

$$a_{b_D} \sum_{i < \delta_0 D} \binom{D}{i} \le \mathrm{poly}(D) e^{-D[\mathsf{h}(1/4) - \mathsf{h}(\delta_0)]},$$

where $\mathsf{h}(t) = -t\log t - (1-t)\log(1-t)$. The entropy gap is positive. Finally $\tau_r \ge cD^{-\beta}$ by Lemma 3.2, proving (24). □

Lemma 8.2 (Sparse adaptive tail Gram matrices)**.** *Fix $A \ge 1$, a failure probability $\delta > 0$, and an odd integer $r_0 \ge 11$. There is $\alpha_0 > 0$ such that, for any fixed $0 < \alpha \le \alpha_0$, the following assertions hold uniformly as $d \to \infty$. Suppose r is odd,*

$$r_0 \le r \le \alpha d, \qquad AD_r \log(D_r + 1) \le n < AD_{r+2}\log(D_{r+2}+1).$$

Put $u = r+2$, $\rho = r/(d+1)$, and $H = \log(e/\rho)$. For each such specified n, with probability at least $1-\delta$,

$$\begin{aligned} \|(Q_r)_{SS}\|_{\mathrm{op}} &\le C_{L,\delta}\left[1 + \frac{\sqrt{nd\, s_{2,r}}}{\tau_r} + \frac{n\ell_r}{\tau_r}\right]\tau_r \\ &\le C_{L,\delta}[1 + \sqrt{A\rho^{-1}H} + AH]\tau_r, \qquad (26) \\ \|(Q_u)_{SS} - \tau_u I_n\|_{\mathrm{op}} &\le [C_{L,\delta}(\sqrt{A\rho H} + A\rho^2 H) + o(1)]\tau_r. \qquad (27) \end{aligned}$$

The $o(1)$ is uniform over the stated range. In particular, after reducing α_0 if necessary, $\|(Q_u)_{SS}\|_{\mathrm{op}} \le 2\tau_r$ on the same event for all sufficiently large d. Deterministically,

$$\begin{aligned} 0 \le \tau_r - \tau_u &\le C_L \tau_r / r, \qquad (28) \\ n\ell_u/\tau_r &\le C_L A\rho^2 H, \qquad (29) \end{aligned}$$

where $\ell_v = \lambda_{v+1}$. No simultaneous event over different sample sizes is asserted.

Proof. The dimension and spectral estimates already proved give

$$n \le CA\binom{D}{r+2} rH, \qquad \log(2n) \le C_A rH,$$
$$n\ell_r/\tau_r \le C_L AH, \qquad nd\, s_{2,r}/\tau_r^2 \le C_L A\rho^{-1}H.$$

For $u = r+2$, the exact two-step binomial ratio gives

$$n\ell_u/\tau_r \le C_L A\rho^2 H, \qquad nd\, s_{2,u}/\tau_r^2 \le C_L A\rho H.$$

Moreover $\tau_r - \tau_u = \binom{D}{r+2} a_{r+2}$, so the level and trace estimates prove (28).

We apply the proof of Lemma 7.2 with a different deterministic mask. Retain pairs satisfying

$$\left|\frac{1 + x \cdot y}{d+1}\right| \le a_n, \qquad a_n = \sqrt{128 \log(2n)/d}.$$

By choosing α_0 sufficiently small, uniformly $\log(2n)/d \le C_A \alpha \log(e/\alpha) + o(1)$ is as small as needed, so $a_n \le a_0$. For independent uniform inputs, Hoeffding's bound (and $a_n \ge 2/(d+1)$) gives exceptional-pair probability

$$b_n \le 2\exp(-a_n^2 d/8) = 2(2n)^{-16}.$$

The exceptional set is translation invariant on the cube. Its size is therefore the same in every row, so the masked-matrix proof applies with $B_{\mathrm{near}}/M = b_n$. This mask is deterministic for each fixed pair (d, n); it does not depend on the observed sample. Lemma 8.1 supplies the retained-entry bound

$$\frac{\eta_v}{\tau_v} \le \left[C\sqrt{\frac{\log(2n) + v + 2}{d}}\right]^{v+2} + Cd^{\beta} e^{-cd}, \qquad v \in \{r, u\}. \tag{30}$$

Thus exactly the argument of Lemma 7.2, still with $q = 2\log M = 2d\log 2$, gives for every $z > 0$

$$\begin{aligned}&\mathbb{P}\{\|(Q_v)_{SS} - \tau_v I_n\| > z\tau_v\} \\ &\quad \le \frac{C}{z^2}\left[\frac{q\eta_v}{\tau_v} + \frac{\sqrt{nqs_{2,v}}}{\tau_v} + \frac{n\ell_v}{\tau_v} + n(M^{-1} + b_n)\right]^2 + n^2(M^{-1} + b_n).\end{aligned} \tag{31}$$

The collision and exceptional-pair terms tend to zero uniformly: $n \ge AD_{r_0}\log(D_{r_0}+1) \to \infty$, and α_0 can be chosen so that $n \le e^{c_1 d}$ with $2c_1 < \log 2$.

The coherence term $q\eta_v/\tau_v$ also tends to zero uniformly. For $r \le \sqrt{d}$, its bracket in (30) is at most $C_A d^{-1/4}\sqrt{\log d}$, and its exponent is at least $r_0 + 2 \ge 13$. Hence it is bounded, after multiplication by q, by

$$Cd[C_A d^{-1/4}\sqrt{\log d}]^{13} + Cd^{1+\beta}e^{-cd} = o(1).$$

For $r > \sqrt{d}$, reducing α_0 makes the bracket at most $1/2$, so the bound is $Cd\,2^{-\sqrt{d}} + Cd^{1+\beta}e^{-cd} = o(1)$. Using (31) for $v = r, u$, with failure probability at most $\delta/2$ for each, now proves (26) and (27). For the latter, multiply by $\tau_u/\tau_r \le 1$ before using the spectral bounds, so no lower comparison of the two traces is needed. Finally $\rho H \to 0$ as $\rho \to 0$, uniformly for $0 < \rho \le \alpha$; choose α_0 to make the fixed part of the fine deviation below $1/2$, and absorb the uniform $o(1)$. Since $\tau_u \le \tau_r$, this gives the asserted upper bound $2\tau_r$. □

9 The complete low-degree head

9.1 Squared balance on the target coordinates

Lemma 9.1 (Active-coordinate squared balance)**.** *If P has total Walsh degree at most r, then*

$$\mathbb{E}(fP)_-^2 \ge \left(\frac{1}{2} - \sqrt{\frac{r}{k}}\right)\|P\|_2^2. \tag{32}$$

For every fixed $\delta > 0$, there is a universal A_δ such that if $r \le k/64$ and $n \ge A_\delta D_r \log(D_r + 1)$, then, with probability at least $1 - \delta$,

$$\mathbb{P}_n(fP)_-^2 \ge \frac{1}{4}\|P\|_2^2 \quad \textit{for every polynomial } P \textit{ of degree at most } r. \tag{33}$$

Proof. For $j \in U$ put $D_j P = (P - P \circ \mathrm{flip}_j)/2$. Because f changes sign under these flips, the inequality $|F(s) - F(t)| \le (|s| + |t|)|s - t|$ for $F(t) = t|t|$ gives

$$|\mathbb{E} f P |P|| \le 2 \|P\|_2 \|D_j P\|_2.$$

Averaging over $j \in U$ and using $\sum_{j \in U} \|D_j P\|_2^2 \le r \|P\|_2^2$ proves (32) from $(fP)_-^2 = (P^2 - fP|P|)/2$. The vector of all Walsh characters of degree at most r is isotropic and has squared length D_r at every input. Under $r \le k/64$, the population bound is at least $3\|P\|_2^2/8$. Apply Lemma 6.3 at accuracy $1/8$ and use homogeneity to obtain (33). □

9.2 Separating the first omitted pair of Fourier levels

Lemma 9.2 (The first paired shell)**.** *Fix $A \ge 1$, $\delta > 0$, and an odd $r_0 \ge 11$. There is a sufficiently small $\alpha > 0$ such that, uniformly for all large d, the following holds. Suppose r is odd and*

$$r_0 \le r \le \alpha d, \qquad A D_r \log(D_r + 1) \le n < A D_{r+2} \log(D_{r+2} + 1).$$

Write $\rho = r/D$, $H = \log(e/\rho)$, and $\tau = \tau_r$. Then, with probability at least $1 - 2\delta$,

$$\|(Q_r)_{SS}\|_{\mathrm{op}} \le C_{L,A,\delta} H \tau. \tag{34}$$

Deterministically,

$$n\lambda_{r+1} \le C_L A H \tau, \tag{35}$$

$$n\lambda_{r+3} \le C_L A \rho^2 H \tau. \tag{36}$$

Proof. Let $B_r = K_{(r,r+2]}$, so $Q_r = B_r + Q_{r+2}$. The two levels in B_r have the common normalized eigenvalue $\ell = \lambda_{r+1} = \lambda_{r+2}$ and total dimension

$$m = \binom{d}{r+1} + \binom{d}{r+2} = \binom{D}{r+2}.$$

Its diagonal is $\Delta = m\ell = \tau_r - \tau_{r+2}$. Choose the feature vector $v(x) = \sqrt{\ell}(\chi_V(x))_{r<|V|\le r+2} \in \mathbb{R}^m$. Then $\|v(x)\|_2^2 = \Delta$ and $\mathbb{E} v v^{\mathsf{T}} = \ell I_m$. For $A_i = v(X_i)v(X_i)^{\mathsf{T}} - \ell I_m$,

$$\|A_i\|_{\mathrm{op}} \le \Delta, \qquad \left\| \sum_{i=1}^n \mathbb{E} A_i^2 \right\|_{\mathrm{op}} \le n\ell\Delta.$$

The self-adjoint matrix Bernstein inequality [4, Theorem 1.4], with $h = \log(2m/\delta)$, therefore gives, with probability at least $1 - \delta$,

$$\|(B_r)_{SS}\|_{\mathrm{op}} \le n\ell + C\sqrt{n\ell\Delta h} + C\Delta h. \tag{37}$$

Here the nonzero eigenvalues of the sample Gram matrix equal those of $\sum_i v(X_i)v(X_i)^{\mathsf{T}}$.

The sharp spectrum and maximality interval imply

$$n \le CAmrH, \quad \ell \le \frac{C_L \tau}{rm}, \quad \Delta \le \frac{C_L \tau}{r}, \quad h \le C_\delta r H.$$

Thus every term in (37) is at most $C_{L,A,\delta}\tau H$. The fine-tail conclusion of Lemma 8.2 for the current adaptive cutoff r gives $\|(Q_{r+2})_{SS}\| \le 2\tau$ with probability at least $1 - \delta$ after reducing α and increasing d_0. The triangle inequality proves (34). The first eigenvalue estimate is among the displayed spectral bounds; the two-step binomial ratio $\binom{D}{r+2}/\binom{D}{r+4} \le C\rho^2$ proves the second. No independence between the two sample events is used. □

9.3 The complete head at an exact sample size

Proposition 9.3 (Head norm and population leakage)**.** *Let r be odd, $r \le k/64$, $\rho = r/(d+1)$, $H = \log(e/\rho)$, and $n = n_r$ from (2). Outside an event of arbitrarily small fixed probability, every minimizer of (3), with $P = z_{\le r}$, satisfies*

$$\|z\|_K^2 \le Cna^2 d^\beta, \tag{38}$$

$$\|P\|_2^2 \le Ca^2 \rho^{-\beta} H, \tag{39}$$

$$\|z - P\|_2^2 \le Ca^2 \rho^{2-1/400-\beta} H. \tag{40}$$

The estimates hold simultaneously for all minimizers.

Proof. The interpolation comparator gives (38); at the displayed sample size, its d term is absorbed by nd^β. The one-sided squared sampling theorem and the parity balance estimate give

$$\frac{1}{4}\|P\|_2^2 \le \frac{1}{n}\sum_i (f(X_i)P(X_i))_-^2 \le \frac{1}{n}\sum_i (z(X_i) - P(X_i))^2 \le \frac{\|z\|_K^2}{n}\|(Q_r)_{SS}\|.$$

Lemma 9.2 bounds the last Gram matrix by $CH\tau_r$, proving (39). The first omitted pair has common eigenvalue $\lambda_{r+1} = \lambda_{r+2}$, dimension $\binom{d+1}{r+2} \asymp D_r \rho^{-2}$, and trace at most $C\tau_r/r$. Therefore

$$n_r \lambda_{r+1} \le C\tau_r H \rho^{2-1/400}.$$

Since $\|z - P\|_2^2 \le \lambda_{r+1}\|z\|_K^2$, this proves (40). The sample size belongs to the interval in Lemma 9.2, because $1/400 < 2$. □

In particular the low-degree head inherits the sign of z outside a set of vanishing population measure, and its mass in $(-a/2, a/2)$ tends to zero. The estimate (39) concerns the entire head; the much weaker bound $\lambda_0\|z\|_K^2$ would not suffice below.

10 Uniform middle-band estimates by Gaussian perturbation

We first fix notation for the geometric estimates in this and the next sections. After relabeling coordinates, write $x = (u, v)$ with $u \in \{-1, 1\}^k$ on the parity support and $v \in \{-1, 1\}^m$ on the remaining coordinates, where $m = d - k$. Write $X_i = (U_i, V_i)$ for the training inputs. The nuisance sphere $S_h(v)$ consists of the points at Hamming distance h from v; S_h also denotes uniform averaging on that sphere. For a permutation π of the nuisance coordinates, $C_{\pi,q}(x)$ is the subcube obtained by freeing its first q coordinates, and $C_{\pi,q}$ denotes the corresponding averaging operator. Put $\mu_q = \mathbb{E}_\pi C_{\pi,q}$. An active face is obtained by freeing a specified set of active coordinates and fixing all others. Its dimension is denoted by m_f. A contact is a second active input on the same face, coupled to its center; the particular coupling used here is defined in Section 15.

Choose an odd cutoff $v_* = C_0 r + O(1)$ and write $z = P + J + T$, where $P = z_{\le r}$, $J = z_{(r,v_*]}$, and $T = z_{>v_*}$ are Fourier projections. The constant C_0 will be fixed in the fine-tail estimate of Section 14. This section uses the exact sample size (2). Put $\tau = \tau_r$, $\rho = r/(d+1)$, and $H = \log(e/\rho)$. Let $B = Cna^2 d^\beta$ be a deterministic upper energy bound. We use the depth-uniform spectral bounds of Section 4; constants may depend on the fixed polynomial sparsity window and a fixed upper multiple v_*/r. Throughout the applications, $\epsilon \ge \rho^e$ for a sufficiently small fixed $e > 0$; the positive-power remainders are asserted only in this range. Auxiliary targets have uniform marginal, or marginal density bounded by a fixed constant relative to uniform. Their source is uniform, and the coupling is candidate independent until the finite-net extension at the end of the section.

Lemma 10.1 (A rare-event Gaussian PAC bound)**.** *Let $\mathcal{H}$ be finite dimensional and $F_h(X) = \max_{j\le M}\|T_{X,j}h\|$ for linear maps into Hilbert spaces. Suppose*

$$\operatorname{tr}(T_{X,j}^* T_{X,j}) \le \Delta, \qquad \|T_{X,j}^* T_{X,j}\| \le L, \qquad \mathbb{P}\{F_h > z\} \le \Lambda\|h\|^2/z^2.$$

Choose a centered isotropic Gaussian prior with variance σ^2 such that $\mathbb{P}_G\{F_G > z/4\} \le \varepsilon_0$ uniformly in X. Put $p_* = 16\Lambda B/z^2 + \varepsilon_0$ and suppose $p_* \le e^{-8}$. With probability at least $1-\delta$, simultaneously for all $\|h\|^2 \le B$,

$$\mathbb{P}_n\{F_h > z\} \le C\sqrt{p_*} + C\frac{B/(n\sigma^2) + \log(2\sqrt{n}/\delta)/n}{\log(1/p_*)}.$$

The conclusion remains true for auxiliary-averaged losses, with the original n independent observations as the sampling units.

Proof. For the bounded loss $\ell_g = 1_{\{F_g > z/2\}}$, the binary PAC-Bayes inequality holds simultaneously for every posterior Q:

$$\operatorname{kl}(\mathbb{E}_Q\mathbb{P}_n\ell_g \,\|\, \mathbb{E}_Q\mathbb{P}\ell_g) \le \frac{\mathrm{KL}(Q\|\Pi) + \log(2\sqrt{n}/\delta)}{n}.$$

For completeness, the binomial exponential-moment bound $\mathbb{E}\exp[n\operatorname{kl}(\mathbb{P}_n\ell\|\mathbb{P}\ell)] \le 2\sqrt{n}$, Fubini, Markov, and the entropy variational inequality prove this; joint convexity of binary relative entropy passes from individual losses to Q. The same argument applies to bounded losses in $[0,1]$ by Bernoulli randomization and Jensen. Choose $Q = N(h, \sigma^2 I)$, so $\mathrm{KL}(Q\|\Pi) = \|h\|^2/(2\sigma^2)$. The triangle inequality gives empirical posterior loss at least $(1-\varepsilon_0)\mathbb{P}_n(F_h > z)$ and population posterior loss at most p_*. If its empirical value q exceeds $\sqrt{p_*}$, then

$$\operatorname{kl}(q\|p') \ge q\log(1/p_*) - q\log(e/q) \ge \tfrac{1}{4}q\log(1/p_*) \qquad (p' \le p_*).$$

Otherwise $q \le \sqrt{p_*}$. These alternatives prove the result. Pointwise Gaussian estimates can be averaged over auxiliary variables before applying the same bounded-loss inequality. □

The needed perturbation probabilities follow from Gaussian norm concentration. In particular,

$$\sigma^2 = \frac{z^2}{C\Delta} \implies \varepsilon_0 \le Me^{-c\Delta/L}, \qquad \sigma^2 = \frac{z^2}{C\Delta D_0}, \quad D_0 \ge \log(2M) \implies \varepsilon_0 \le e^{-cD_0}. \tag{41}$$

Indeed the mean norm is at most $\sigma\sqrt{\Delta}$ and the squared Lipschitz constant is at most $\sigma^2 L$.

10.1 Dyadic completed pairs

Index omitted pairs by $j \ge 1$, with pair j containing degrees $r+2j-1, r+2j$. Group $t \le j < 2t$ for $t = 1, 2, 4, \ldots$, stopping at v_*. If I_t denotes a block, its kernel trace Δ_t and largest eigenvalue Λ_t satisfy

$$\Delta_t \le C(t/r)\tau, \qquad B\Lambda_t/a^2 \le CH\rho^{-\beta-1/400}(C\rho)^{2t}. \tag{42}$$

These follow by summing the paired level masses and iterating their two-step binomial ratio. Thus the first omitted pair is included. Give block $t = 2^s$ threshold $a\epsilon_t$, where $\epsilon_t = c\epsilon/(s+1)^2$ and $\sum_t \epsilon_t \le \epsilon$. The triangle inequality reduces a bad event for the sum to the union of block bad events. All fields may be sample dependent.

10.2 Nested angular fluctuations

For centered restriction to a nuisance subcube, the feature covariance in a block has trace at most Δ_t and norm at most $C\rho\Delta_t$. To verify the norm bound, diagonalize on the free coordinates. Every nonconstant restricted character contains some specified free coordinate; the sum of original Fourier weights containing that coordinate is at most $(v_*/d)\Delta_t$. This bounds each restricted eigenvalue. For each fixed nuisance order, reverse-martingale Doob gives

$$\mathbb{P}_X\left\{\max_q \mathrm{Var}_{C_{\pi,q}(X)}\, h > z^2\right\} \le \|h\|_2^2/z^2 \le \Lambda_t\|h\|_K^2/z^2.$$

The same bound holds after averaging random orders. Use the first choice in (41), with at most $d+1$ stages. The perturbation failure is $e^{-c/\rho}$, and (42) gives $\log(1/p_*) \ge c\min(tH, 1/\rho)$. Lemma 10.1 and dyadic summation yield

$$\mathbb{P}_{n,\pi}\left\{\max_q \mathrm{Var}_{C_{\pi,q}(X)}\, J > a^2\epsilon^2\right\} \le C\epsilon^{-2}\left[\frac{\rho^{-\beta}\log^5(r+2)}{rH} + \rho^{1-\beta}\log^4(r+2)\right] + \rho^c, \tag{43}$$

with negligible confidence terms at the sample sizes under consideration. Replacing ϵ^2 by $p_{\mathrm{bin}}\epsilon^2$ divides the displayed complexity terms by p_{bin}. This controls subcube angular fluctuations; it asserts no spherical weak-type estimate.

10.3 Local common offsets

Let Y differ from X only on an active face of dimension m_f. For $h \le S$, the feature map $h_0 \mapsto [h_0(X) - h_0(Y,W)]_{W\in S_h(V)}$ has squared Hilbert–Schmidt norm at most

$$C\rho(m_f + S)\Delta_t.$$

The bound follows by averaging character differences: a degree at most v_* support meets the changed coordinates with probability at most $C\rho(m_f+S)$. For the population mean-square maximum, the crude union bound gives

$$\mathbb{E}\max_{h\le S}\mathbb{E}_{W\in S_h(V)}|h_0(X) - h_0(Y,W)|^2 \le C(S+1)\Lambda_t\|h_0\|_K^2.$$

Only the source and target marginals are used. Choose $D_0 = C[tH + \log(2S+2)]$ in the second Gaussian option. If $S \le \rho^{-\Theta}$ and $2 - 1/400 - \beta - \Theta > 0$, the rare-event logarithm is at least ctH. Consequently

$$\mathbb{P}_{n,Y}\left\{\max_{h\le S}\mathbb{E}_{W\in S_h(V)}|J(X) - J(Y,W)|^2 > a^2\epsilon^2\right\} \le C\epsilon^{-2}\rho^{1-\beta}(m_f+S)\log^4(r+2) + \rho^c. \tag{44}$$

This proves the mean-square version needed to preserve a global gap.

10.4 Radial windows and steps

Let $R_b^m(h)$ be the normalized degree-b nuisance sphere multiplier. Its difference identity is

$$R_b^m(h) - R_b^m(h+1) = \frac{2b}{m}R_{b-1}^{m-1}(h).$$

The Krawtchouk decay estimate implies, for $q \ge 128\ell$, $|h - q/2| \le \sqrt{2q\ell}$, and $b \le v_*$,

$$|R_b^m(h) - R_b^m(\lfloor q/2\rfloor)|^2 \le C\ell b/m.$$

Indeed sum the difference identity across the window and use $|R_{b-1}^{m-1}(h)| \le C\exp[-c(b-1)q/m]$; maximizing $(b/m)^2 q\ell \exp[-c(b-1)q/m]$ proves the claim, with $b = 0, 1$ handled directly. Window feature variance is therefore at most $C\ell\rho\Delta_t$. The dimension-free spherical-mean maximal theorem [6] gives

$$\left\| \max_h |S_h h_0| \right\|_2 \le C\|h_0\|_2,$$

so it bounds the population loss of radial windows. Gaussian union over at most d^2 windows and the preceding dyadic argument give

$$\mathbb{P}_n \left\{ \max_{\substack{q \ge 128\ell \\ |h-q/2| \le \sqrt{2q\ell}}} |S_h J - S_{\lfloor q/2 \rfloor} J| > a\epsilon \right\} \le C\epsilon^{-2}\rho^{1-\beta}\ell \log^4(r+2) + \rho^c. \tag{45}$$

This invokes the maximal theorem for ordinary means, not for angular variances or for spherical averages of squared functions.

For permutation-averaged subcube means the multiplier is $A_b(q) = \binom{m-q}{b}/\binom{m}{b}$, with $A_b(q) - A_b(q+1) = bA_b(q)/(m-q)$. Its absolute value is at most Cb/m for $b \le v_*$. The step feature variance is thus at most $C\rho^2\Delta_t$. Jensen and reverse Doob give the population L^2 maximal bound for μ_q. The same proof yields

$$\mathbb{P}_n\{\max_q |\mu_q J - \mu_{q+1} J| > a\epsilon\} \le C\epsilon^{-2}\rho^{2-\beta} \log^4(r+2) + \rho^c. \tag{46}$$

10.5 Fixed couplings and a finite net

Every proof above remains valid when the target active point is drawn from a fixed candidate-independent coupling whose source is uniform, whose target marginal density is bounded by a fixed constant, and whose active changes stay within the indicated face. The pointwise Gaussian feature bounds are unchanged, while population bounds gain only the marginal-density constant. For a finite set of N_* such couplings, apply each PAC bound with confidence δ/N_*. The additional complexity is $\log N_*/n$, divided by the same rare-event logarithm. A data-dependent head coupling is handled by the finite pattern net in Section 15; no covariance concentration event is unioned over that net.

11 Spherical and permutation covariance envelopes

The spherical angular estimate requires a genuine maximal covariance bound. A maximal theorem for ordinary spherical means alone would not control averages of the squared field.

Lemma 11.1 (Intrinsic positive-matrix concentration)**.** *Let independent positive matrices A_i satisfy $A_i \le LI$. Put $\mu = \|\sum_i \mathbb{E}A_i\|$ and $T = \operatorname{tr}\sum_i \mathbb{E}A_i$. With probability at least $1-\delta$,*

$$\left\| \sum_i A_i \right\| \le C\left[\mu + L\log\left(1 + \frac{T}{L\delta}\right)\right].$$

Proof. The matrix Laplace inequality and $e^{A_i/L} \le I + (e-1)A_i/L$ give, with $M = \sum_i \mathbb{E}A_i$,

$$\mathbb{E}\operatorname{tr}(e^{\sum_i A_i/L} - I) \le \operatorname{tr}(e^{(e-1)M/L} - I) \le (e-1)(T/L)e^{(e-1)\mu/L}.$$

Since the left random trace is at least $e^{\|\sum_i A_i\|/L} - 1$, Markov proves the result. Subtracting the identity before Markov removes the ambient dimension. □

11.1 A homogeneous spherical envelope

On an m-cube give every degree-b character eigenvalue κ and write $\pi_b = \binom{m}{b}\kappa$. Let C_h be the centered covariance of its evaluation feature on the sphere of radius h about the all-plus point.

Lemma 11.2 (Homogeneous envelope)**.** *For $m \geq 5$ there is a positive operator $E_b \geq C_h$ for every h, with*

$$\|E_b\| \leq \frac{\pi_b}{m-1}, \qquad \operatorname{tr} E_b \leq (b+1)\pi_b.$$

If $b \leq c\sqrt{m}$, then $\operatorname{tr} E_b \leq C_c\sqrt{b+1}\,\pi_b$.

Proof. The degree-b subset module decomposes multiplicity freely into harmonic components indexed by $0 \leq j \leq \min(b, m-b)$, of dimensions $M_j = \binom{m}{j} - \binom{m}{j-1}$. One elementary construction starts with the kernels of the subset down operator and lifts them by the up operator; the relation $DU - UD = (m-2j)I$ on degree j gives the direct sum and the stated dimensions. Permutation-invariant covariances are scalar on these components. Sphere centering removes $j = 0$; every remaining component has dimension at least $m-1$. Take the maximum covariance eigenvalue over h separately on each component. Since every C_h has trace at most π_b, this proves the first two bounds.

For the sharper trace, a representative in component j is

$$\prod_{i=1}^{j}(x_{2i-1} - x_{2i}) \sum_{\substack{|B|=b-j \\ B\subseteq\{2j+1,\dots,m\}}} \chi_B(x).$$

Its uniform squared norm is $2^j\binom{m-2j}{b-j}$. On sphere h each difference pair must contain exactly one minus sign, an event of probability $2^j(h)_j(m-h)_j/(m)_{2j}$. Consequently the fraction of homogeneous trace in component j is

$$T_j(h) = M_j \frac{4^j(b)_j(m-b)_j(h)_j(m-h)_j}{(m)_{2j}^2}\left[R_{b-j}^{m-2j}(h-j)\right]^2. \tag{47}$$

Here $(x)_j$ is a falling factorial; impossible sphere parameters give zero. Thus $\operatorname{tr} E_b/\pi_b = \sum_{j\geq 1}\max_h T_j(h)$.

Write normalized Krawtchouk as a polynomial in the population mean $t = 1 - 2h/n$. Its monic version $P_l = (n)_l R_l/n^l$ satisfies

$$P_{l+1}(t) = tP_l(t) - \frac{l(n-l+1)}{n^2}P_{l-1}(t).$$

The associated Jacobi matrix has symmetric real roots of absolute value at most $2\sqrt{l/n}$. Pairing opposite roots and bounding the leading coefficient therefore gives

$$|R_l^n(t)|^2 \leq e^{Cl^2/n}(t^2 + 4l/n)^l \qquad (l \leq n/2).$$

Set $s = 4(h/m)(1-h/m)$. Falling-factorial estimates in (47), followed by this root bound, give uniformly for $b \leq c\sqrt{m}$

$$T_j(h) \leq e^{Cb^2/m}\binom{b}{j}s^j(1 - s + Cb/m)^{b-j} \leq C_c\mathbb{P}\{\mathrm{Bin}(b, s/(1+Cb/m)) = j\}.$$

Stirling's bounds imply $\sup_t \mathbb{P}\{\mathrm{Bin}(b,t) = j\} \leq C\sqrt{(b+1)/[(j+1)(b-j+1)]}$. Summing over j proves the square-root trace bound. □

11.2 The complete highpass covariance

Let $q \geq q_0$ be odd, with q_0 a sufficiently large fixed constant, $q \leq d/8$, and $k \leq d^{1-\eta}$. Put $m = d - k$ and $\tau = \tau_q$.

Proposition 11.3 (Maximal spherical angular covariance)**.** *There is a deterministic positive operator A_X on the RKHS of Q_q such that, for every field t,*

$$\max_h \operatorname{Var}_{w \in S_h(V)} t(U, w) \leq \langle t, A_X t \rangle, \quad \|A_X\| \leq C\tau/d, \quad \operatorname{tr} A_X \leq C\tau d^2.$$

Its translation mean has norm at most $Cq^2\lambda_{q+1}$. If $q \leq c_0\sqrt{d-k}$ for a sufficiently small fixed c_0, this improves to $Cq^{3/2}\lambda_{q+1}$. Hence in that range, with probability $1-\delta$, simultaneously for all t,

$$\sum_i \max_h \operatorname{Var}_{S_h(V_i)} t(U_i, \cdot) \leq C\left[nq^{3/2}\lambda_{q+1} + \frac{\tau}{d}\log(1 + nd^3/\delta)\right]\|t\|_K^2.$$

The same statement holds with an independent uniform active input averaged inside every summand.

Proof. At a fixed active input u, collapse the active Fourier coordinates. For each nuisance support B, $b = |B|$, the corresponding mutually orthogonal feature vectors have squared norm

$$\kappa_b = \sum_a \binom{k}{a} \lambda_{a+b} 1_{a+b>q}, \qquad \pi_b = \binom{m}{b} \kappa_b.$$

Apply Lemma 11.2 separately at each nuisance degree. Put $s = \lfloor q/4 \rfloor$, $\varepsilon = 1/4$, and for $b \geq s$ choose

$$w_b = (b+1)^{-1-\varepsilon} + s^{\beta-\varepsilon}\pi_b, \quad W = \sum_{b \geq s} w_b \leq Cs^{-\varepsilon}.$$

Weighted Cauchy shows that $A_{\mathrm{hi}} = W \bigoplus_{b \geq s}(E_b/w_b)$ dominates the variance of the sum of these homogeneous pieces. Since $\pi_b/w_b \leq s^{\varepsilon-\beta}$, its norm is at most $Cs^{-\beta}/m$ and its trace at most $Cs^{-\beta}m^2$.

Uniform sign translation diagonalizes its mean. Pulling back from the collapsed features to an original support with active degree a and nuisance degree b cancels κ_b: its mean diagonal is at most

$$Cs^{-\varepsilon}(b+1)^{2+\varepsilon}\lambda_{a+b}.$$

The paired spectrum implies, for every fixed $w > 0$,

$$\sup_{j>q}(j+1)^w \lambda_j \leq C_w(q+1)^w \lambda_{q+1}.$$

Up to degree $d/4$, this follows from the growing paired binomial denominators; beyond that point monotonicity gives exponential suppression. Thus the high-piece mean is at most $Cq^2\lambda_{q+1}$. With the sharper homogeneous trace, the power $2+\varepsilon$ becomes $3/2+\varepsilon$ for $b \leq c\sqrt{m}$. The remaining larger degrees are exponentially suppressed if $q \leq c_0\sqrt{m}$ with c_0 small, giving $Cq^{3/2}\lambda_{q+1}$.

The omitted low nuisance degrees have total trace $\Delta_{\mathrm{lo}} \leq \exp[-cq\log(d/k)]$: conditional on total degree $j > q$, fewer than $q/4$ nuisance coordinates require more than $3j/4$ active coordinates, with probability at most $2^j(k/d)^{3j/4}$. The equal-weight envelope $s\bigoplus_{1 \leq b < s} E_b$ has norm at most $Cq\Delta_{\mathrm{lo}}/d$, negligible relative to τ/d, and mean at most $Cq^2\lambda_{q+1}$, or $Cq^{3/2}\lambda_{q+1}$ in the dilute range. Degree zero has no angular variance. Combining the two pieces and applying Lemma 11.1 proves the result. □

11.3 Permutation means

For a translation-invariant spectral subkernel of trace τ and largest normalized eigenvalue λ, and a nuisance permutation π, let $L_{X,\pi,q}$ be the feature averaged on the first-q free-coordinate cell and $\overline{L}_{X,q} = \mathbb{E}_\pi L_{X,\pi,q}$. Put $J_d = 1 + \lceil \log_2(d+1) \rceil$. There is a positive envelope A_X^{perm} satisfying

$$\begin{aligned} \mathbb{E}_\pi \max_q |\langle h, L_{X,\pi,q} - \overline{L}_{X,q} \rangle|^2 &\le \langle h, A_X^{\mathrm{perm}} h \rangle, \\ \|A_X^{\mathrm{perm}}\| \le C J_d^2 \tau/d, \qquad & \operatorname{tr} A_X^{\mathrm{perm}} \le C J_d^2 \tau, \\ \|\mathbb{E} A_X^{\mathrm{perm}}\| &\le C J_d^2 \lambda. \end{aligned}$$

To construct it, express each prefix as a sum of at most J_d disjoint dyadic intervals and apply Cauchy. On a fixed dyadic scale, interval feature differences select disjoint Walsh support families, so total trace is at most τ. Centering over permutations removes the trivial representation. The one-dimensional sign representation is absent: a dyadic interval feature depends only on an ordered partition into three coordinate blocks; for $m > 3$ its stabilizer contains a transposition. Every remaining irreducible component has dimension at least $m - 1$, giving the norm gain. Translation averaging is diagonal; for each support the disjoint interval indicators have total variance at most one per scale, giving the mean bound.

For source inputs, Lemma 11.1 now gives a uniform empirical bound of order $J_d^2[n\lambda + \tau \log(nd/\delta)/d]$. For a contact selected from the finite head-pattern net, use Gaussian PAC rather than unioning this matrix event. In a dyadic band the envelope has trace $C J_d^2 \Delta_t$, norm smaller by d, and population mean $C J_d^2 \Lambda_t$. Gaussian perturbation with variance $z^2/(C J_d^2 \Delta_t)$ has uniform noise failure e^{-cd}. Lemma 10.1 and dyadic summation therefore give

$$\begin{aligned} &\mathbb{P}_{n,\mathrm{contact}} \left\{ (\mathbb{E}_\pi \max_q |C_{\pi,q} J - \mu_q J|^2)^{1/2} > a\epsilon \right\} \\ &\qquad \le C\epsilon^{-2} \frac{\rho^{-\beta} \log^2 d \log^5(r+2)}{rH} + \rho^c + \frac{C \log N_* \log(r+2)}{nH}. \end{aligned}$$

This is a positive power of ρ in the large-r branch $r \ge \rho^{-4/5}$, for the small threshold powers used below.

11.4 Coupled contacts in the small-cutoff branch

The source event in Proposition 11.3 cannot be unioned over the head-pattern net. Instead compare a contact U' with the original active input U. For a middle-band block I_t, the feature map

$$h_0 \longmapsto [h_0(U', W) - h_0(U, W)]_{W \in S_h(V)}$$

has trace at most $C\rho m_f \Delta_t$, uniformly at every radius: only the at most m_f changed active coordinates enter the character difference. The full mean-square maximal difference has population bound

$$\mathbb{E} \max_h \mathbb{E}_{W \in S_h(V)} |h_0(U', W) - h_0(U, W)|^2 \le C r^{3/2} \Lambda_t \|h_0\|_K^2.$$

Indeed bound each endpoint's second moment by its variance plus squared mean, then use the dilute envelope and the ordinary HKS maximal theorem. The contact's bounded marginal density changes only the constant. Gaussian norm concentration, a union over at most $d + 1$ radii, and Lemma 10.1

therefore give

$$\mathbb{P}_{n,\mathrm{contact}}\left\{\max_h \mathbb{E}_{S_h(V)}|J(U',W)-J(U,W)|^2 > a^2\epsilon^2 p_{\mathrm{bin}}\right\} \le C\epsilon^{-2}\frac{\rho^{1-\beta}m_f\log^4(r+2)}{p_{\mathrm{bin}}} + \rho^c + \frac{C\log N_*\,\mathrm{polylog}(r)}{n}.$$

Here $r \le \rho^{-4/5}$ and $p_{\mathrm{bin}} = \rho^{\beta/2+30e}$, $\epsilon = \rho^e$. The first block's population exponent is at least $2 - 1/400 - 3\beta/2 - 6/5 - 32e > 0$; the complexity exponent is at least $1 - 3\beta/2 - 32e > 0$. Finally,

$$\mathrm{Var}_h\, J(U',\cdot) \le 2\,\mathrm{Var}_h\, J(U,\cdot) + 2\mathbb{E}_{S_h(V)}|J(U',W)-J(U,W)|^2.$$

The source full-tail envelope, together with the uniform fine-tail angular event, bounds the first term. The fine-tail event also holds at the contact. This supplies the required full-tail spherical angular control at adaptive balanced contacts without an invalid matrix union.

12 Radial feature bounds and endpoint anchors

Throughout this section the nuisance dimension $m = d - k$ is comparable to d. The bounds also hold after a positive spectral restriction.

Lemma 12.1 (Uniform Krawtchouk decay)**.** *For normalized Krawtchouk polynomials,*

$$|R_s^n(h)| \le \exp\left[-c\frac{\min(s,n-s)\min(h,n-h)}{n}\right].$$

Proof. Reflection of either index preserves absolute value, and normalized duality gives $R_s(h) = R_h(s)$. It suffices to treat $R_b(a)$ with $1 \le a \le b \le n/2$. Pair $q = \max(1, \lfloor a/4 \rfloor)$ negative population signs with positive signs. In the average over uniform b-subsets, any subset splitting a pair cancels with the subset obtained by switching membership within its first split pair. Hence the absolute average is at most the probability that no pair splits. For $a \ge 4$, expose pairs successively. Conditional on previous non-splitting patterns, at least $b/2$ selected and $n/4$ unselected positions remain. Each next pair splits with probability at least $b/(4n)$. Since $q \ge a/7$, the bound follows. For $a = 1, 2, 3$, a single pair splits with probability at least b/n, which gives the same bound after reducing the absolute constant. □

Lemma 12.2 (Sphere increments and windows)**.** *Let L_h be the kernel feature averaged on nuisance sphere h. Then, uniformly for $0 \le h < m$,*

$$\|L_h - L_{h+1}\|_K^2 \le Cd^{-\beta}[1 + \min(h, m-1-h)]^{\beta-2}.$$

Consequently, if $q \ge 128\ell$, $q \le m$, and $|h - q/2| \le \sqrt{2q\ell}$, then

$$\|L_h - L_{\lfloor q/2 \rfloor}\|_K^2 \le Cd^{-\beta}q^{\beta-1}\ell.$$

For an active modification of distance l,

$$\|K_{(u,v)} - L_h(u',v)\|_K^2 \le Cd^{-\beta}(l+h+1)^{\beta}.$$

Proof. Let $\pi_j = \binom{m}{j}\kappa_j$ be the nuisance spectral distribution obtained by fixing the active input. Its degree is no greater than the full spectral degree, so $\sum_{j\ge t}\pi_j \le C(t+1)^{-\beta}$. Tail integration gives $\sum_j \pi_j j^2 e^{-tj} \le Ct^{\beta-2}$ for $0 < t \le 1$. Monotonicity of κ_j also gives $\sum_{j>3m/4}\pi_j \le 2e^{-cm}$. The exact difference formula yields

$$\|L_h - L_{h+1}\|_K^2 = \frac{4}{m^2}\sum_{j=1}^{m}\pi_j j^2 [R_{j-1}^{m-1}(h)]^2.$$

By reflection assume $h \le (m-1)/2$. For $j \le m/2$, apply Lemma 12.1 and the exponential second-moment bound. For $m/2 < j \le 3m/4$, the Krawtchouk factor is at most Ce^{-ch} and the spectral mass at most $Cm^{-\beta}$. The remaining degree range is exponentially small. This proves the first claim, including $h = 0$ by the unweighted bound $\mathbb{E}j^2 \le Cm^{2-\beta}$. Telescoping feature norms across the central window proves the second. The third follows from the endpoint cusp bound $1 - K(x,y) \le C(\mathrm{dist}(x,y)/d)^{\beta}$ and Jensen. □

12.1 Midpoint and full-nuisance means

For the total-degree highpass Q_r, the squared feature norm of nuisance midpoint averaging is at most

$$\Delta_{\mathrm{mid}} \le C\tau_r(k/d)^{r+1} + C\tau_r/d^2 + C\sqrt{d}\,2^{-(d-k)}. \tag{48}$$

To verify this, expand by nuisance degree b. At the middle sphere, $|R_b^m(\lfloor m/2\rfloor)| \le C/m$ for $1 \le b < m$. For even $m = 2t$ this follows directly from $(1-z)^t(1+z)^t = (1-z^2)^t$; for odd $m = 2t+1$ the generating function is $(1+z)(1-z^2)^t$. Divide coefficients by $\binom{m}{b}$ and use reflection to bound the remaining ratios by their first internal value. The $b = 0$ mass requires an entirely active support of degree above r, giving the first term. The $b = m$ mass is exponentially small by spectral monotonicity. Full nuisance averaging retains only $b = 0$ and therefore has just the first term.

Uniform translation makes the population endpoint operator norm at most λ_{r+1}. For source inputs, Lemma 11.1 applies directly. For a fixed contact coupling with bounded target marginal, use a scalar Gaussian prior of variance $a^2\epsilon^2/(C\Delta_{\mathrm{mid}}H)$ instead. Its uniform perturbation failure is e^{-cH}; the shifted KL divided by n is at most $C\epsilon^{-2}d^{\beta}\Delta_{\mathrm{mid}}H$. Lemma 10.1 gives coupled endpoint bad fraction

$$\rho^c + C\epsilon^{-2}d^{\beta}\Delta_{\mathrm{mid}} + \frac{C[\log N_* + \log(n/\delta)]}{nH}$$

simultaneously for a finite net of N_* couplings. Polynomial sparsity makes the pure-active term negligible for growing r; the d^{-2} term is harmless. This avoids a union of endpoint matrix concentration events over head-dependent contacts.

13 Head recurrence and local drift

Lemma 13.1 (Uniform recurrence of a head window)**.** *Let P range over a D_0-dimensional linear space. Partition a bounded range into J_0 fixed intervals. For a center and an independent nuisance order, call it bad if its interval has relative mass less than q in some nested nuisance cell. For every $p_0 > 0$, the empirical bad fraction is, simultaneously for all P, at most*

$$Cq/p_0 + CJ_0p_0 + C(d+1)e^{-Lq/16} + C\sqrt{\frac{D_0\log(2 + D_0J_0(d+1)(L+1)) + \log(1/\delta)}{n}},$$

with probability $1-\delta$*, provided* $Lq \geq 10$*. The loss may be averaged over nuisance orders and auxiliary active inputs. For canonical head-dependent face contacts, the same bound has an additional* $O(D_0 m_f)$ *complexity and a fixed marginal-density factor.*

Proof. For a fixed head and fixed active input, ignore intervals of population mass less than p_0, losing at most $J_0 p_0$. If A is another interval, under the law conditioned on membership in A, the reciprocal of its conditional likelihood is a nonnegative supermartingale. Ville's inequality shows that its cell mass drops below q on at most a $q/\mathbb{P}(A)$ fraction of this conditional law. Summing gives the fixed-head bound $q/p_0 + J_0 p_0$.

For uniformity, draw L independent ghost points in every nested cell. Use the event that some cell has at most $3Lq/2$ ghost points in the center's interval. A truly bad center triggers this ghost event with probability at least $1/2$. If all cell masses exceed $2q$, the ghost event has probability at most $(d+1)e^{-Lq/16}$ by Chernoff. Its fixed-head expectation is therefore bounded by $2q/p_0 + J_0 p_0$ plus this term. For N augmented observations, all ghost labels use only $O(NJ_0(d+1)(L+1))$ affine comparisons in the head coefficients. The arrangement bound gives VC dimension $CD_0 \log(2+D_0 J_0(d+1)(L+1))$. VC entropy, symmetrization, and Jensen for ghost averaging prove the stated uniform bound.

For a canonical contact, first fix its face sign/order pattern. The chosen contact is then fixed for each auxiliary random seed and all ghost head evaluations remain linear in the coefficients. The additional $O(4^{m_f})$ pattern comparisons add $O(D_0 m_f)$ to the VC dimension. Population bad mass grows by at most the contact marginal-density constant. This is a bounded head-only argument; it makes no covariance-event union over contact patterns. □

Lemma 13.2 (Maximal local head drift)**.** *For degree at most* r *and* $S \leq (d-k)/2$*, the population local drift is bounded by*

$$\mathbb{E}_X \max_{h \leq S} \mathbb{E}_{W \in S_h(V)} |P(U,W) - P(U,V)|^2 \leq C \frac{rS}{d-k} \log^2(2S) \|P\|_2^2.$$

The same bound holds empirically, uniformly over the degree-r *space, up to a fixed constant when* $n \geq CD_r \log(D_r/\delta)$*. For head-dependent balanced contacts, the threshold bad fraction is controlled by this population bound and bounded VC transfer.*

Proof. Couple the spheres by a uniformly random distinct-coordinate nuisance path. Every path prefix is a union of at most $O(\log S)$ dyadic blocks. Cauchy bounds the maximum squared increment by $O(\log S)$ times the sum of squared dyadic block increments. A block of length l changes a degree at most r character with probability at most $rl/(d-k)$. Translation orthogonality therefore bounds the expected block quadratic form by $Clr/(d-k)$ times the identity. On each dyadic scale the total length is at most $2S$, proving the population estimate. The resulting positive envelope has mean norm $\mu = C(rS/(d-k)) \log^2(2S)$ and trace at most $D_r \mu$. The trace bound is pointwise (translation changes character signs), so the envelope norm is at most $D_r \mu$. Matrix Chernoff gives its empirical bound for the stated sample size. For a balanced contact, the population bound gains only its fixed target marginal density. The indicator of a threshold failure is determined, on a face pattern cell, by at most $S+1$ quadratic forms in the head coefficients. Its VC dimension is $CD_r[m_f + \log(2S) + 1]$, so bounded statistical transfer applies. This avoids multiplying a matrix bound by the number of head patterns. □

The head cap used in recurrence is justified similarly by $\|P\|_2^2 \leq Ca^2 \rho^{-\beta} H$ and bounded threshold VC transfer. Taking cap $a\rho^{-\beta/2-10e}$, interval width at most $a/16$, $p_0 = \rho^{\beta/2+20e}$, and $q = \rho^{\beta/2+30e}$ makes the uniform recurrence failure a positive power of ρ at the exact sample sizes. Markov over the auxiliary nuisance order then gives simultaneous recurrence along at least three quarters of orders outside a positive-power exceptional set of center/contact pairs.

14 Transport of the all-input margin

Write an input as (u, v), with $u \in \{-1, 1\}^k$ active and $v \in \{-1, 1\}^m$ nuisance, where $m = d - k$. For a nuisance function, let S_h denote averaging on the Hamming sphere of radius h about its indicated center. For a permutation π of the nuisance coordinates, let $C_{\pi,q}$ average over the subcube obtained by freeing the first q coordinates, and put $\mu_q = \mathbb{E}_\pi C_{\pi,q}$. Choose an odd $v_* = C_0 r + O(1)$ and split

$$z = P + J + T, \qquad P = z_{\le r}, \quad J = z_{(r,v_*]}, \quad T = z_{>v_*}.$$

The letter v_* denotes a cutoff, not a nuisance input.

Lemma 14.1 (Effective profiles)**.** *Fix $\eta > 0$ and the exact sample sizes of Theorem 1.1. There are constants $g, \Theta > 0$ such that the following holds with probability tending to one, simultaneously for all globally separated, training-compatible fields satisfying (38) and (39). Put $S = \lfloor \rho^{-\Theta} \rfloor$. Choose a random active face of dimension m_f, where m_f is bounded by a fixed power of H, through the observed active input U_i. For a uniform auxiliary point, or a canonical balanced contact u from Section 15 on an eligible face, define*

$$W_{i,u,h} = P(u, V_i) + J(X_i) + S_h T(u, V_i), \qquad 0 \le h \le 2S.$$

Outside an empirical fraction at most ρ^g of center–face–point pairs, the profile has margin $a(1 - o(1))$ at every radius, and

$$\operatorname{sign} W_{i,u,S} = \operatorname{sign} P(u, V_i).$$

The same margin conclusion holds at the distinguished input $u = U_i$ outside a fraction ρ^g of centers; at radius zero its value equals $z(X_i)$ exactly. The statement concerns auxiliary averages, rather than a maximum over arbitrary active points.

14.1 Uniform high-frequency witnesses

Lemma 14.2 (Fine-tail evaluation and contrast Gram matrices)**.** *Let $v_* = C_0 r + O(1)$, with C_0 a sufficiently large fixed constant. Assume $\log n = O(rH)$, $k = O(rH)$, and $\log d = O(H)$. At every center allow all active translations and a polynomial number of nuisance sphere averages, adjacent increments, and radial window contrasts. Give each scalar contrast a deterministic variance upper bound at least an inverse polynomial in d, and normalize by that bound. With probability $1 - \exp(-crH)$, every adaptive choice of one such normalized contrast per center has Gram norm at most a fixed constant. Moreover,*

$$\sum_i \sup_{u,h} \operatorname{Var}_{w \in S_h(V_i)} T(u, w) \le C \frac{\tau_{v_*}}{d} \|T\|_K^2.$$

Proof. For translated, uniformly bounded signed evaluation averages, Fourier orthogonality bounds the second moment of a cross-center covariance by $C\lambda_{v_*+1}\tau_{v_*}$. The analogous Hilbert–Schmidt bound holds for the centered spherical evaluation operators. Their individual operator norms are at most $C\tau_{v_*}/(d - k - 1)$: nuisance permutation invariance removes the trivial representation, and each remaining irreducible component has dimension at least $d - k - 1$. A union over the at most $2^k \operatorname{poly}(d)$ choices at each center, followed by Markov and Gershgorin, bounds all normalized cross-entries by $1/n$. Its failure probability is at most

$$Cn^4 4^k \operatorname{poly}(d) \lambda_{v_*+1} \tau_{v_*}.$$

The spectral entropy bound makes this at most $\exp(-crH)$ after increasing C_0. The block Hilbert–Schmidt argument gives the second assertion. All choices may be made after observing the field. □

14.2 Proof of the effective-profile lemma

Proof of Lemma 14.1. Choose $e = 1/40000$, $\epsilon = \rho^e$, $p_{\mathrm{bin}} = \rho^{\beta/2+30e}$, and $S = \lfloor \rho^{-(1-\beta)/2} \rfloor$. Lemma 13.2, the head norm bound, and (44) give, outside positive-power fractions of source/contact pairs, simultaneously for $h \le 2S$,

$$\mathbb{E}_{S_h(V_i)} |P(u,W) - P(u,V_i)|^2 \le a^2\epsilon^2, \quad \mathbb{E}_{S_h(V_i)} |J(u,W) - J(X_i)|^2 \le a^2\epsilon^2.$$

The fine-tail angular Gram bound gives the same smallness for the fluctuation of T about its sphere mean. On each sphere Chebyshev leaves a point where all deviations are $O(a\epsilon)$; the all-input margin of z therefore gives $|W_{i,u,h}| \ge a(1 - O(\epsilon))$. At the observed input and radius zero the decomposition is exact.

We now anchor the sign. Lemma 13.1 supplies, for at least three quarters of nuisance orders, a head-value window of width at most $a/16$, containing $P_0 = P(u,V_i)$, with mass at least p_{bin} in every nested cell. The statement holds uniformly for the original head and for its canonical contacts. The cap failure, ghost recurrence error, and contact-pattern mismatch all have a positive power bound at the exact sample sizes.

Suppose first that $r \le \rho^{-4/5}$. The dilute envelope and its contact extension give, outside a positive-power exceptional fraction,

$$\max_h \mathrm{Var}_{S_h(V_i)}(J+T)(u,\cdot) \le a^2 p_{\mathrm{bin}} \epsilon^2/100.$$

Indeed after multiplying the envelope by the energy budget and dividing by $p_{\mathrm{bin}}\epsilon^2$, its coherent term has exponent at least $2 - 1/400 - 3\beta/2 - 6/5 - 32e > 0$; its diagonal term has a larger exponent. Average the recurrent window mass over nuisance permutations. A uniformly freed q-subcube is exactly a mixture of spheres with radius $Z \sim \mathrm{Bin}(q, 1/2)$. Put $\ell = \log(100/p_{\mathrm{bin}})$. At least a fixed multiple of p_{bin} of its joint mass lies in the matching head window and in $|Z - q/2| \le \sqrt{2q\ell}$. Angular Chebyshev cannot remove this entire mass. At a surviving point the global margin, the head-window width, and the middle/fine radial-window bounds yield

$$|P_0 + S_{\lfloor q/2 \rfloor}(J+T)(u,V_i)| \ge 3a/4 \qquad (2S \le q \le m).$$

The radial-window estimate (45) controls J; Lemma 12.2 and the uniform fine-tail Gram event control T. Their fine-tail bad fraction is at most $C\epsilon^{-2}S^{\beta-1}\ell$, with no division by p_{bin}. Adjacent radial steps are small, so these gapped expressions cannot change sign. The midpoint endpoint is small by (48) and its scalar-PAC contact extension. The common sign is therefore $\operatorname{sign} P_0$. At radius S, the local middle-band approximation transfers this sign to $W_{i,u,S}$.

For $r \ge \rho^{-4/5}$ use the hybrid profile

$$H_q = P_0 + \mu_q J(u,V_i) + S_{\lfloor q/2 \rfloor} T(u,V_i), \qquad 2S \le q \le m.$$

Equation (43), at threshold $a^2 p_{\mathrm{bin}}\epsilon^2$, controls the middle-band subcube variance. Its leading exponents are at least $4/5 - 3\beta/2 - 32e$ and $1 - 3\beta/2 - 32e$, both positive. The permutation-mean estimate controls the difference from $\mu_q J$. For source inputs use its intrinsic matrix bound; for balanced contacts use its Gaussian-PAC finite-net form. Markov over orders makes both properties simultaneous along at least three quarters of orders.

Their intersection with the recurrent-window orders has mass at least one half. In each good cell delete the middle deviation set, costing at most $p_{\mathrm{bin}}/100$. Averaging the remaining matching mass over orders leaves at least $0.49 p_{\mathrm{bin}}$ in the binomial sphere mixture. The binomial tail outside the central window costs at most $p_{\mathrm{bin}}/100$. The fine-tail spherical angular event, which is uniform over

all active contacts, removes at most another $p_{\mathrm{bin}}/100$. At a surviving point all three approximations hold. The fine radial-window bound thus gives $|H_q| \geq 3a/4$ for every q. This step uses subcube angular control only for J and spherical angular control only for T.

The binomial step estimate (46) and the fine sphere-step Gram bound make consecutive H_q differ by $o(a)$. The full-nuisance mean of J and the midpoint mean of T are small by the endpoint bounds, including their contact-PAC form. Hence every H_q has sign $\operatorname{sign} P_0$. Convex averaging of the local middle-band mean-square estimate gives $|\mu_{2S} J(u, V_i) - J(X_i)| \leq a\epsilon$. Therefore $W_{i,u,S}$ has the same sign.

All error estimates above are powers of ρ times fixed logarithms. The smallest branch slack at $\beta = 1/2$ is $0.05 - 1/400 - 32e > 0$. Recurrence errors and pattern mismatches are also positive powers; Markov over auxiliary orders and the finitely many uses reduces these exponents only by fixed factors. Taking, for example, $g = 5e$ after all reductions is valid. This proves the lemma, uniformly in the field, including balanced contacts. □

14.3 The two-scale witness

Lemma 14.3 (A wrong head value yields a witness)**.** *Let a face have dimension m_f, and normalize its head by two lower same-sign amplitude quantiles. Write $Q = (P - t)/s$, with $s \geq a/2$, and let F be the projection of Q onto the halfline of* $\operatorname{sign} P$ *with absolute margin one. Fix an amplitude cap R with $8(m_f + 1) \leq R \leq S$. At a wrong center outside the quantile exception set put $b = |F|$. Suppose a same-head-sign quantile contact has an effective profile as in Lemma 14.1. One adaptive fine-tail scalar witness then has squared value divided by its variance upper bound at least*

$$ca^2 d^\beta \begin{cases} b^2/(m_f + b + 1)^\beta, & 1 \leq b \leq R/4, \\ R^{2-\beta}, & b > R/4. \end{cases}$$

Proof. The head normalization gives a center–contact contrast at least cab whenever the contact profile has its original sign. If $b \leq R/4$, inspect radius $h_0 = \max(1, \lceil b \rceil)$; otherwise inspect $h_0 = \lfloor R/2 \rfloor$. If the original sign is already present, use the center–contact contrast, whose variance is at most $Cd^{-\beta}(m_f + h_0 + 1)^\beta$. Otherwise the sign anchor at S forces an adjacent sign-changing pair at some radius $h \geq h_0$. Its jump has magnitude at least ca, while its variance is at most $Cd^{-\beta}(h_0 + 1)^{\beta-2}$. These two alternatives give the displayed bounds. The common scalar offset cancels in both contrasts. □

The adjacent-shell variance estimate follows from the exact Krawtchouk multiplier formula and the cusp exponent β:

$$\operatorname{Var}_K(S_h - S_{h+1}) \leq Cd^{-\beta}(h + 1)^{\beta-2}.$$

The center–contact estimate follows from the kernel increment bound $1 - K(x, y) \leq C(\operatorname{dist}(x, y)/d)^\beta$ and Jensen. Lemma 14.2 adds one chosen witness per center. Consequently, for $p = 2 - \beta$, the empirical head-only losses satisfy

$$\mathbb{P}_{n,\mathrm{face}}[1_{\mathrm{wrong}} b^p 1_{b \leq R/4}] \leq C(m_f + 1)^\beta + CR^p \delta_{\mathrm{geom}} + 1, \tag{49}$$

$$\mathbb{P}_{n,\mathrm{face}}[1_{\mathrm{wrong}} 1_{b > R/4}] \leq CR^{-p} + \delta_{\mathrm{geom}}. \tag{50}$$

The geometric bad fraction obeys $\delta_{\mathrm{geom}} \leq \rho^g$ after quantile contact selection. The cap $R = \rho^{-t_{\mathrm{rad}}}$ is chosen much smaller than the anchor horizon S. This separation ensures that $R^p \delta_{\mathrm{geom}} \to 0$.

15 Balanced contacts and the sharper face obstruction

The maximum possible distance between two points on an active face is its dimension. A more useful comparison preserves the distribution of the center and gives the contact a bounded density. The resulting distance has moments on the square-root scale. We first prove the coupling statement, including the simultaneous tail bound needed below.

Lemma 15.1 (A bounded-density coupling)**.** *Let μ be uniform on $\{-1,1\}^m$, and let ν have density at most $C \geq 1$ relative to μ. There is a single coupling $X \sim \mu$, $Y \sim \nu$ such that, for every $t \geq 0$,*

$$\mathbb{P}\{\operatorname{dist}(X,Y) > 2\sqrt{m(\log C + 2t)} + t\} \leq 2e^{-t}. \tag{51}$$

Consequently, for every $q \geq 1$,

$$\|\operatorname{dist}(X,Y)\|_q \leq C_0\sqrt{m(\log C + q)} + C_0 q.$$

Two measures of densities at most C_1, C_2 can likewise be coupled with distance L satisfying

$$\|L\|_q \leq C_0\sqrt{m(\log C_1 + \log C_2 + q)} + C_0 q. \tag{52}$$

Proof. Expose coordinates in a fixed order and write

$$p_i(y_{<i}) = \nu\{Y_i = 1 \mid Y_{<i} = y_{<i}\}, \quad \delta_i = |p_i - 1/2|, \quad S_i = \sum_{j \leq i} \delta_j^2.$$

On a zero-probability prefix the conditional probability may be defined arbitrarily. Let L_i be the likelihood ratio of the first i coordinates under ν against their uniform law. Under the uniform law, conditional on a positive-likelihood prefix,

$$\mathbb{E}[\sqrt{L_i} \mid Y_{<i}] = \sqrt{L_{i-1}}\,\frac{\sqrt{p_i} + \sqrt{1-p_i}}{\sqrt{2}}.$$

The square of the last factor is

$$\frac{1 + \sqrt{1 - 4\delta_i^2}}{2} \leq 1 - \delta_i^2 \leq e^{-\delta_i^2}.$$

Thus $\sqrt{L_i}e^{S_i/2}$ is a nonnegative supermartingale. Since $L_m \leq C$,

$$\mathbb{E}_\nu e^{S_m/2} = \mathbb{E}_\mu L_m e^{S_m/2} \leq \sqrt{C}\,\mathbb{E}_\mu \sqrt{L_m} e^{S_m/2} \leq \sqrt{C}. \tag{53}$$

Given the full joint past, maximally couple a fresh uniform Bernoulli X_i with a Bernoulli variable Y_i of parameter $p_i(Y_{<i})$. The conditional marginal of X_i is uniform even after conditioning on the full past; hence X is uniform, and Y has law ν. The disagreement indicator has conditional mean δ_i. For $D = \operatorname{dist}(X,Y)$ and $A = \sum_i \delta_i \leq \sqrt{mS_m}$, the Bernoulli exponential supermartingale gives

$$\mathbb{E}\exp\{D - (e-1)A\} \leq 1.$$

Because $e - 1 < 2$, the event $D > 2\sqrt{mS_m} + t$ has probability at most e^{-t}. Equation (53) also gives $\mathbb{P}\{S_m > \log C + 2t\} \leq e^{-t}$. Their union proves (51); integration proves the moment bound. Finally, glue the two couplings through a common uniform intermediate point and use the triangle inequality and Minkowski's inequality. □

Fix a sufficiently small positive constant ζ_0. Write μ for uniform probability on the face and set $\operatorname{sign} 0 = 1$. A face is eligible when each head-sign class $S_\sigma = \{x : \operatorname{sign} P(x) = \sigma\}$ has mass at least $2\zeta_0$ and $\mu\{|P| < a/2\} < \zeta_0/2$. For $\sigma \in \{-1, 1\}$ define the lower fixed-mass quantile

$$b_\sigma = \min\{b \geq 0 : \mu(S_\sigma \cap \{|P| \leq b\}) \geq \zeta_0\}.$$

All points tied at the quantile value are included. The strict lower set has mass less than ζ_0, so ties cause no ambiguity in the projection estimate below. Eligibility implies $b_\sigma \geq a/2$ and the usable contact sets

$$A_\sigma = S_\sigma \cap \{a/2 \leq |P| \leq b_\sigma\} \quad \text{satisfy} \quad \mu(A_\sigma) \geq \zeta_0/2.$$

Put

$$s = (b_+ + b_-)/2 \geq a/2, \qquad t = (b_+ - b_-)/2, \qquad Q = (P - t)/s.$$

Project $Q(x)$ onto the halfline $\{y : (\operatorname{sign} P(x))y \geq 1\}$ and call the resulting value $F(x)$. Then $|F| \geq 1$ and $\operatorname{sign} F = \operatorname{sign} P$. A displacement occurs only on the strict lower quantile set in one of the two sign classes. There $Q \in [-1, 1]$, so the displacement is at most two and

$$\|F - Q\|_2^2 \leq 8\zeta_0. \tag{54}$$

On ineligible faces all statistical losses are defined as zero. Apply Lemma 15.1 to the uniform laws on S_σ and A_σ, and mix the two couplings with weights $\mu(S_\sigma)$. This defines a joint law of a uniform center U and a contact V. The contact has the same original head sign, its marginal density is at most $2/\zeta_0$, and

$$\|\operatorname{dist}(U, V) + 1\|_q \leq C_{\zeta_0}\{\sqrt{mq} + q + 1\} \qquad (q \geq 1). \tag{55}$$

Fix the coordinate order and all one-bit coupling conventions once. The resulting contact law depends only on the sign and ordering pattern of the head on the face. This observation permits uniform control when the head itself depends on the training data.

15.1 Uniformity for a head-dependent contact law

Lemma 15.2 (A finite net of face patterns)**.** *Let heads range over a D_0-dimensional real linear space. Let $c_P(\mathcal{F})$ record the signs, orderings, and fixed threshold comparisons on an m-face that determine eligibility and the contact sets. For $0 < \epsilon < 1/2$, there is a data-independent finite set of heads $P_1, \ldots, P_M$ with*

$$\log M \leq CD_0\{m + \log(1/\epsilon) + 1\} \tag{56}$$

such that every P has an index j with $\mathbb{P}_\mathcal{F}\{c_P(\mathcal{F}) \neq c_{P_j}(\mathcal{F})\} \leq \epsilon$. With probability at least $1 - \delta$ over n independent uniform training inputs, the same index has empirical disagreement, averaged over faces through those inputs, at most

$$\epsilon + C\sqrt{\frac{D_0(m + 1) + \log(1/\delta)}{n}}. \tag{57}$$

Proof. Each face pattern is determined by at most $C4^m$ affine comparisons in the head coefficients. On N faces, the disagreement indicators of a pair of heads are therefore determined by an arrangement of $CN4^m$ hyperplanes in $2D_0$ parameters. The number of cells, including cells on hyperplanes, is at most $(CN4^m/D_0)^{CD_0}$ for $N \geq D_0$. Solving the shattering inequality shows that the disagreement class has VC dimension at most $CD_0(m + 1)$; an extra $\log D_0$ is unnecessary here.

The VC epsilon-net theorem supplies a fixed auxiliary collection of at most $CD_0(m+1)\epsilon^{-1}\log(2/\epsilon)$ faces meeting every disagreement set of probability greater than ϵ. Partition coefficient space by the

comparisons on those faces and choose a representative in each nonempty cell. The arrangement bound gives (56). Two heads in one cell agree on the auxiliary collection, so their population disagreement is at most ϵ. The auxiliary collection establishes existence of the net and is independent of the training data.

The VC entropy bound gives uniform empirical deviation $C\sqrt{(D_0(m+1)+\log(1/\delta))/n}$ for the disagreement class. Jensen's inequality preserves this bound when the auxiliary face is averaged before evaluating each training input. This proves (57). □

For each fixed net index, the contact law is an ordinary fixed augmented-input distribution. Its source is uniform and its target has bounded density. The Gaussian perturbation estimates used in the middle-band arguments are uniform in the augmented observation, and their population estimates use only these marginals. Applying those arguments simultaneously to the net costs $\log M$ in the PAC-Bayes numerator. On an agreeing face the entire conditional contact law agrees, so the discarded fraction (57) transfers the conclusion to the data-dependent head. At the specified sample sizes, $D_0 = D_r$, $m \le CrH$, and $\epsilon = \rho^c$, both this fraction and $\log M/n$ are positive powers of ρ, up to logarithmic factors.

The marginal statement alone would not justify this substitution: a coupling can concentrate on a source/contact relation having small product measure. The finite-pattern argument supplies the required uniformity for that relation. Likewise, one must not union a D_r-dimensional matrix concentration bound over all net indices. Head-only drift and recurrence losses instead use their bounded VC estimates; the middle-band extensions use Gaussian PAC-Bayes.

15.2 Weighted witnesses and bounded statistical losses

Write $\ell = \operatorname{dist}(U, V)$, and let $b = |F(U)|$ for the projected, contact-normalized head. Use separate scales: the effective radial profile is anchored at a horizon S, whereas amplitudes are capped at $R \ll S$, with $m \ll R$. At a good wrong center outside the contact-quantile exception, the proof of the two-scale witness lemma with the actual distance ℓ gives a normalized scalar witness of squared value at least

$$ca^2 d^\beta \begin{cases} b^2/(\ell+b+1)^\beta, & b \le R/4, \\ R^{2-\beta}, & b > R/4. \end{cases} \tag{58}$$

For completeness, test the contact profile at $h_0 = \lceil b \rceil$ in the first case and at $h_0 = \lfloor R/2 \rfloor$ in the second. If it has the original head sign, compare it with the observed center. Otherwise an adjacent sign change between h_0 and S supplies the witness. The contrast and increment variances are bounded respectively by $Cd^{-\beta}(\ell+h+1)^\beta$ and $Cd^{-\beta}(h+1)^{\beta-2}$. This proves (58) while allowing all intermediate sign relays.

Since $p = 2-\beta$ and $b \ge 1$,

$$\frac{b^2}{(\ell+b+1)^\beta} \ge c\frac{b^p}{(\ell+1)^\beta}.$$

The uniform fine-tail Gram event permits one adaptively chosen scalar witness per training center. Sum that deterministic inequality and then average over the face and the canonical contact law. The comparator budget and a geometric bad-pair fraction δ_g give

$$\widehat{\mathbb{E}}\left[1_{\text{wrong}}\frac{|F(U)|^p}{(\ell+1)^\beta}1_{|F(U)|\le R/4}\right] \le C + CR^p\delta_g, \tag{59}$$

$$\widehat{\mathbb{E}}[1_{\text{wrong}}1_{|F(U)|>R/4}] \le CR^{-p} + \delta_g. \tag{60}$$

Here $\widehat{\mathbb{E}}$ includes all auxiliary averages. The contact-quantile exception has $|F| = 1$, so its empirical contribution is at most one; no empirical smallness is inferred from its population quantile alone.

These are bounded head-only losses after the geometric inequalities have been established. For a fixed contact random seed, the contact and its distance are fixed within each face sign/order cell. A subgraph test reduces to an affine normalized-amplitude comparison after multiplying by the positive normalization scale. Thus the same arrangement and VC-entropy argument gives dimension at most $CD_r(m + \log(D_r + 2))$. Auxiliary averaging preserves the bound. Choose $R = \rho^{-t_0}$ after obtaining $\delta_g \le \rho^g$, with $pt_0 < \min(g, 1/800)/2$. Uniform transfer of (59)–(60) then gives a bounded population weighted cost and a positive-power bound on the mass of huge wrong amplitudes. Only these bounded head-only losses are transferred; the geometric bad relation is not.

15.3 Passing to an accurate face

The contact quantile ζ_0 is a fixed sufficiently small constant. Projection to the prescribed sign half-lines changes the normalized head by at most $C\sqrt{\zeta_0} + o(1)$ in L^2; the fixed term need only lie below the tolerance in Lemma 16.1. The random-restriction leakage still tends to zero. A full-head error at most 3/8 implies that a positive fraction of faces have error at most 7/16. On such a face both sign classes have positive constant mass. Markov selection for the bounded weighted cost and the leakage therefore supplies an eligible accurate face with cost bounded by a fixed constant.

The huge wrong-amplitude set has positive-power mass by (60) and bounded statistical transfer. If the restricted polynomial has degree $u \le a_1 H$, with a_1 sufficiently small, hypercontractivity gives precisely the exceptional-set absorption hypothesis of Lemma 16.1. Its contact-distance hypothesis is (55). The uniform calibration in Section 18 applies when $\beta \log m$ stays bounded. Its constants are independent of depth.

16 Analytic calibration of contact-weighted errors

Let μ_m be uniform probability measure on $\{-1, 1\}^m$. Write

$$f(x) = \prod_{i=1}^{m} x_i, \qquad \partial_i P(x) = \frac{P(x) - P(x^{\oplus i})}{2}, \qquad \Delta = \sum_{i=1}^{m} \partial_i.$$

Thus $\Delta \chi_S = |S| \chi_S$. All unqualified norms and expectations use μ_m. For $1 < t < 2$, put

$$s_t = 1 + \frac{2}{\pi} \arcsin\left(\frac{2}{t} - 1\right).$$

We use the dimension-free scalar Bernstein inequality

$$\|\Delta P\|_t \le C_t u^{s_t} \|P\|_t \qquad (\deg P \le u). \tag{61}$$

This is part of the established theory of polynomial inequalities on the cube [5]. Section 17 supplies the needed version, including locally uniform dependence on t, from the imaginary-power estimate of Carbonaro and Dragičević [7].

Fix $0 < \beta \le 1/2$ and put $p = 2 - \beta$. A contact law is a joint distribution π of (U, V) in the cube whose first marginal is exactly μ_m. Set $L = \operatorname{dist}(U, V)$. For a margin-separated function F, meaning $|F| \ge 1$, define its parity-error set

$$E_F = \{x : f(x)F(x) < 0\}.$$

The contact-weighted error cost outside an exceptional set B is

$$\mathcal{C}_{\pi,B}(F) = \mathbb{E}_\pi \frac{\mathbf{1}_{\{U \in E_F \setminus B\}} |F(U)|^p}{(L+1)^\beta}. \tag{62}$$

No independence between the amplitudes, labels, and contact distance is assumed.

Lemma 16.1 (Stable contact-cost calibration)**.** *Fix $1 < p_0 < p$, $c_0 > 0$, $K < \infty$, and $C_* < \infty$. Suppose that:*

1. *$|F| \geq 1$ and $\mathbb{E}[f \operatorname{sign} F] \geq c_0$;*
2. *Q has degree at most $u \geq 1$ and $\|F - Q\|_{p_0} \leq e \leq c_0/2$;*
3. *$\mu_m(B) \leq \delta$ and*
$$\delta^{1/p_0 - 1/2}(p_0 - 1)^{-u/2} \leq \frac{1}{16}; \tag{63}$$
4. *$\mathcal{C}_{\pi,B}(F) \leq K$ and, for every $q \geq 1$,*
$$\|L+1\|_{L^q(\pi)} \leq C_*(\sqrt{mq} + q + 1). \tag{64}$$

Then

$$u \geq c\, m^{\theta(p_0)}, \qquad \theta(p_0) = \frac{1 - \beta/(2p)}{s_{p_0}}, \tag{65}$$

where $c > 0$ depends only on the displayed fixed parameters.

The exceptional set may depend arbitrarily on F, Q and the contact law. Its small probability is not sufficient on its own; condition (63) is what controls its amplitude contribution. In particular, for $\delta = \rho^b$, it suffices to have $u \leq a \log(1/\rho)$ with a sufficiently small.

Theorem 16.2 (Limiting exponent)**.** *Under the hypotheses of Lemma 16.1, choose, for sufficiently large m, $p_0 = p - 1/\log(em)$ and impose (63) uniformly over these exponents. Then*

$$u \geq c\, \frac{m^{\theta_*}}{[\log(em)]^\kappa}, \qquad \theta_* = \frac{1 - \beta/(2p)}{s_p}, \qquad \kappa = \frac{\beta}{2p s_p}. \tag{66}$$

For $\beta = 1/2$, these constants are

$$\theta_* = 0.6851115714\ldots, \qquad \kappa = 0.1370223142\ldots.$$

16.1 Proof of the calibration theorem

We first record two simple consequences of parity.

Lemma 16.3 (Parity and positivity)**.** *If $|F| \geq 1$, $g = \operatorname{sign} F$, and $\mathbb{E}[fg] \geq c_0$, then*

$$\mathbb{E}[g \Delta F] \geq I(g) \geq m c_0, \qquad I(g) = \sum_i \mathbb{P}\{g(X) \neq g(X^{\oplus i})\}. \tag{67}$$

For every real Q and $1 \leq t < \infty$,

$$\|Q\|_t \leq 2\|(fQ)_-\|_t + \frac{2}{m}\|\Delta Q\|_t. \tag{68}$$

Proof. On an edge where g changes sign, margin separation gives $(g(x)-g(x^{\oplus i}))(F(x)-F(x^{\oplus i})) \geq 4$. Edge symmetrization proves the first inequality in (67). Also $\mathbb{E}[fg] = \mathbb{E}[f\partial_i g]$, so $|\mathbb{E}[fg]| \leq \mathbb{E}|\partial_i g| = \mathrm{Inf}_i(g)$ for every i. Summing proves its second inequality; a square on the correlation is unnecessary.

Let A be normalized cube adjacency and $q = fQ$. Since parity reverses on every edge, $(I+A)q = 2f\Delta Q/m$. Write its right side as h. Positivity of A gives $q_+ \leq Aq_- + |h|$. Contractivity of A and the triangle inequality give $\|q\|_t \leq \|q_+\|_t + \|q_-\|_t \leq 2\|q_-\|_t + \|h\|_t$. □

Proof of Lemma 16.1. Set $X = \|Q\|_{p_0}$ and $W = \|F\mathbf{1}_{E_F \setminus B}\|_{p_0}$. Hölder's inequality with conjugate exponents p/p_0 and $p/(p-p_0)$ yields

$$W \leq K^{1/p}\|L+1\|_{L^q(\pi)}^{\beta/p}, \qquad q = \frac{\beta p_0}{p-p_0}. \tag{69}$$

If $q < 1$, its norm is bounded by the first moment. For fixed $p_0 < p$, hypothesis (64) therefore gives

$$W \leq Cm^{\beta/(2p)}. \tag{70}$$

The exact uniform first marginal is used in identifying the left side of (69).

The usual degree-u hypercontractive comparison gives $\|Q\|_2 \leq (p_0-1)^{-u/2}X$. Consequently,

$$\|F\mathbf{1}_B\|_{p_0} \leq \delta^{1/p_0-1/2}\|Q\|_2 + e \leq X/16 + e.$$

Since the negative-part map is 1-Lipschitz, Lemma 16.3 implies

$$X \leq 2W + X/8 + 4e + \frac{2}{m}\|\Delta Q\|_{p_0}. \tag{71}$$

If $2C_{p_0}u^{s_{p_0}}/m \geq 1/4$, then $u \geq cm^{1/s_{p_0}}$, which is stronger than the conclusion. Otherwise (61) absorbs the last term in (71), giving $X \leq C(W+e)$.

Each ∂_i is an L^1 contraction, so $\|\Delta(F-Q)\|_1 \leq me$. By (67),

$$\|\Delta Q\|_{p_0} \geq \mathbb{E}[g\Delta Q] \geq m(c_0 - e) \geq mc_0/2.$$

Combining this with (61), (70), and the bound on X proves (65). □

Proof of Theorem 16.2. Now $q = \beta p_0 \log(em)$ in (69), and hence

$$W \leq Cm^{\beta/(2p)}[\log(em)]^{\beta/(2p)}.$$

The Bernstein constants are bounded on a fixed compact interval of exponents containing all sufficiently large-m values of p_0. Moreover, $s_{p_0} - s_p = O(1/\log(em))$. We may assume $u \leq m$, so $u^{s_{p_0}} \leq Cu^{s_p}$. The preceding proof gives (66). □

17 A logarithmic Bernstein refinement

The next estimate makes the norm dependence in scalar Bernstein explicit. Its proof follows the fixed-vector interpolation method of Domelevo et al. [8], with strip harmonic measure in place of Gaussian damping. We make no priority claim for the interpolation refinement.

Theorem 17.1. *For $1 < t < 2$, set $\sigma_t = (2/\pi)\arcsin(2/t-1)$. Every nonzero real cube polynomial of degree at most $u \geq 1$ satisfies*

$$\|\Delta P\|_t \leq C_t u\left[1 + \log\frac{\|P\|_2}{\|P\|_t}\right]^{\sigma_t}\|P\|_t. \tag{72}$$

The constants are bounded when t ranges over a compact subinterval of $(1,2)$.

Proof. Let $0 < \varepsilon \le (t-1)/t$ and define

$$q = \frac{t(1-\varepsilon)}{1 - t\varepsilon/2}, \qquad \frac{1}{t} = \frac{\varepsilon}{2} + \frac{1-\varepsilon}{q}.$$

We first prove

$$\|\Delta P\|_t \le C_t u \varepsilon^{-\sigma_t} \|P\|_2^{\varepsilon} \|P\|_q^{1-\varepsilon}. \tag{73}$$

For $L = \Delta + \gamma I$, $\gamma > 0$, the imaginary-power estimate of [7, Proposition 11] is

$$\|L^{iv}\|_{q \to q} \le C(q^* - 1)(1 + |v|)^{1/2} e^{\phi_q |v|}, \qquad \phi_q = \arcsin(2/q - 1).$$

Its constants are independent of the dimension and of γ. Fix a real g with $\|g\|_{t'} = 1$, put $N = 1/\varepsilon$, and form the holomorphic bilinear pairing

$$G(z) = \mathbb{E}\left[(L^{N(1-z)} P)\, g_z \right], \qquad g_z = \operatorname{sign}(g) |g|^{t'((1-z)/2 + z/q')}.$$

The value is set to zero where $g = 0$. At $\theta = 1 - \varepsilon$, it equals $\mathbb{E}[(LP)g]$. On the left boundary, $|G(iv)| \le (u + \gamma)^N \|P\|_2$; on the right,

$$|G(1 + iv)| \le C(q^* - 1)(1 + N|v|)^{1/2} e^{\phi_q N |v|} \|P\|_q.$$

On the finite cube G is bounded on the closed strip, so the Poisson inequality applies to $\log |G|$.

The right-boundary harmonic-measure density at θ is

$$w_\varepsilon(v) = \frac{\sin(\pi\varepsilon)}{2(\cosh(\pi v) - \cos(\pi\varepsilon))}, \qquad \int w_\varepsilon = 1 - \varepsilon.$$

The elementary bounds needed here are

$$\int |v| w_\varepsilon(v)\, dv \le \frac{2\varepsilon}{\pi} (\log(1/\varepsilon) + 1),$$
$$\int \log(1 + |v|/\varepsilon) w_\varepsilon(v)\, dv \le C.$$

For the first, expand

$$\frac{\sin a}{\cosh x - \cos a} = 2 \sum_{n \ge 1} \sin(na) e^{-nx}$$

and integrate, obtaining

$$(2/\pi^2) \int_0^{\pi\varepsilon} - \log(2 \sin(s/2))\, ds.$$

For the second, use $w_\varepsilon(v) \le \pi\varepsilon / [4(v^2 + \varepsilon^2)]$ and rescale.

The left boundary has mass ε. Integrating the boundary logarithms consequently gives

$$|G(\theta)| \le C_t (u + \gamma) \|P\|_2^{\varepsilon} \|P\|_q^{1-\varepsilon} \varepsilon^{-2\phi_q/\pi}.$$

Here $q \ge 1 + (t-1)/2$, and $|\phi_q - \phi_t| \le C_t \varepsilon$, so the last power is at most $C_t \varepsilon^{-\sigma_t}$. Taking the dual supremum and then $\gamma \downarrow 0$ proves (73).

Since $q < t$, $\|P\|_q \le \|P\|_t$. Take $\varepsilon = \min\{(t-1)/t, [1 + \log(\|P\|_2 / \|P\|_t)]^{-1}\}$ to obtain (72). All bounds used above are uniform on compact exponent intervals. □

Hypercontractivity gives $\log(\|P\|_2 / \|P\|_t) \le (u/2) \log(1/(t-1))$. Substitution recovers (61), since $s_t = 1 + \sigma_t$. Theorem 17.1 therefore does not prove the general linear-degree Bernstein conjecture; it gives linear degree when the norm ratio is bounded.

18 Uniform linear calibration on logarithmic faces

Lemma 18.1 (Linear face degree at small cusp exponent)**.** *Assume the balanced-contact reduction of the preceding sections, with uniform comparator and witness constants. Suppose $\beta \le 1/4$, $\beta \log m \le B_1$, and $\beta H \to \infty$. An accurate eligible m-face with a polynomial truncation of degree $u \le a_1 H$ must satisfy*

$$u \ge cm,$$

where $a_1 > 0$ is sufficiently small and $c > 0$ depends only on B_1, the fixed contact quantile, the comparator constant, and the fixed accuracy advantage. The constants are independent of depth.

Proof. Put $p = 2 - \beta$ and $p_0 = 2 - 2\beta \in [3/2, 2)$. In the Hölder step of Lemma 16.1, the required distance moment is

$$q = \frac{\beta p_0}{p - p_0} = p_0.$$

The balanced contact distance has uniformly bounded L^{p_0} norm after division by $\sqrt{m}$. Thus the wrong-side L^{p_0} norm is bounded by $Cm^{\beta/(2p)}$, apart from the exceptional-set term. This factor is bounded uniformly when $\beta \log m \le B_1$.

For the exceptional huge-amplitude set, of mass at most $C\rho^{b_0}$ for a fixed exponent $0 < b_0 \le pt_0 < g/2$ (from (50), bounded statistical transfer, and face selection), hypercontractivity gives the absorption factor

$$(p_0 - 1)^{-u/2} (C\rho^{b_0})^{1/p_0 - 1/2}.$$

Its logarithm is at most $-b_0 \beta H/2 + 2\beta u + O(\beta)$ for small β. Choosing $a_1 \le b_0/8$, after t_0 is fixed, makes it tend to $-\infty$ because $\beta H \to \infty$. The fixed quantile projection leakage can be chosen below the accuracy tolerance uniformly, since $p_0 \le 2$.

The scalar Bernstein inequality used in Lemma 16.1 has constants bounded for $p_0 \in [3/2, 2]$. This follows directly from its complex-interpolation proof: the imaginary-power prefactor is bounded on that compact interval, and the angle derivative remains bounded at two. Moreover $s_{p_0} = 1 + O(\beta)$. Consequently $m^{s_{p_0}-1} \le C_{B_1}$ and, for $u \le m$, $u^{s_{p_0}}/m \le C_{B_1} u/m$. The positivity and influence inequalities in the same lemma now imply $mc_0 \le Cu^{s_{p_0}}$ after the fixed leakage is absorbed, with a fixed $c_0 > 0$. Therefore $u \ge cm$. □

19 Random restriction to an accurate face

For a random m_f-face and its degree-u Fourier truncation,

$$\mathbb{E}_{\text{face}} \|P - P_{\text{face},\le u}\|_2^2 \le \|P\|_2^2 \left(\frac{e(r/k)m_f}{u+1} \right)^{u+1}. \tag{74}$$

To see this, expand in Walsh characters. A support of size at most r retains more than u active coordinates with at most the displayed hypergeometric upper-tail probability. Random fixed signs cancel all cross terms. Averaging and summing the squared coefficients proves the inequality.

If the global head error is at most $3/8 + o(1)$, a positive fraction of faces have error at most $7/16$ and hence correlation at least $1/8$. Each sign class on such a face has mass at least $1/16$. The global head leakage and Markov discard only a vanishing fraction for the bad-margin eligibility condition. The population versions of (59)–(60), together with (74), with fixed Markov constants, select one accurate eligible face with controlled wrong-side moment, huge-set mass, and restriction leakage.

Normalize its truncation with the same s, t: $P_0 = (P_{\mathrm{face},\le \mathrm{u}} - t)/s$. The contact projection bound gives

$$\|F - P_0\|_p \le C\sqrt{\zeta_0} + 2\|P - P_{\mathrm{face},\le \mathrm{u}}\|_2/a.$$

Take $u = a_1 H$ with a_1 sufficiently small relative to b_0. The huge-set mass $C\rho^{b_0}$ satisfies the uniform absorption condition of Lemma 18.1. Its conclusion forces $u \ge cm_f$ whenever $\beta \log m_f$ is bounded.

20 Uniform parameter choices and the posterior conclusion

Proof of Theorem 1.1. The depth assumptions imply $\beta^{-1} = O(\log\log d)$, $\beta \log H = O(1)$, and $\beta H \to \infty$ uniformly in the polynomial sparsity window. For any fixed small $\alpha > 0$, an odd $r \asymp \alpha k$ eventually exceeds $\exp(C_1/\beta)$ and satisfies $r \le k/64$. The uniform comparator gives $\|z\|_K^2 \le Cna^2 d^\beta$ for every minimizer. All spectral, Gram, head, and transport bounds consequently have the uniform constants specified in Section 4.

Choose fixed failure probabilities whose sum is at most $1/4$. Suppose a compatible minimizing field has test error below $3/8$. Its complete degree-r head inherits this error and the global margin up to vanishing population errors. Use

$$\zeta_{\mathrm{stat}} = 1/400, \quad e = 1/40000, \quad \epsilon = \rho^e, \quad p_{\mathrm{bin}} = \rho^{\beta/2+30e}, \quad S = \rho^{-(1-\beta)/2}.$$

The head cap is $a\rho^{-\beta/2-10e}$ and the contact quantile mass is a fixed sufficiently small ζ_0. All transport errors are at most ρ^g, with $g = 5e$, uniformly in depth. Choose an amplitude cap $R = \rho^{-t_0}$ after fixing these parameters, with $pt_0 < \min(g, 1/800)/2$. Then the bounded head-only losses transfer to the population with errors tending to zero, and $R \ll S$.

Take $u = \lfloor a_1 H \rfloor$ with $a_1 > 0$ small enough for the uniform bad-set absorption. Set $m_f = \lceil MH \rceil$, with M a fixed constant. The polynomial lower bound on k ensures $m_f \le k$. Choose a_1/M smaller than the constant in Lemma 18.1, and then choose $\alpha = r/k$ so small that $e\alpha m_f/(u+1)$ is a sufficiently small fixed number. The restriction estimate gives

$$\mathbb{E}_{\mathrm{face}}\|P - P_{\mathrm{face},\le \mathrm{u}}\|_2^2 \le Ca^2\rho^{-\beta} H \left(\frac{e\alpha m_f}{u+1}\right)^{u+1} = e^{-cH},$$

after decreasing α: the logarithm of its prefactor is $O(\beta H + \log H) = o(H)$. The balanced-contact and random-face selection therefore provide an accurate eligible face. Since $\beta \log m_f$ is bounded, uniform calibration forces $u \ge cm_f$, contrary to the choice of a_1/M. Thus every compatible minimizer has error at least $3/8$.

Finally, the width limit is taken at each fixed d and its finite $L(d)$. Proposition 2.1 applies without needing any uniform rate in L. Every limiting posterior mixture has draw error at least $3/8$. If F is the fraction of inputs whose correct-label probability is at most $2/3$, then

$$3/8 \le R_{\mathrm{draw}} \le (1-F)/3 + F, \qquad F \ge 1/16.$$

Apply this inequality before the lower width limit to handle arbitrary weights among tied minimizers. The sample events have joint probability at least $3/4$. Since $r \asymp \alpha k$ and $H \asymp \log(ed/k)$, the displayed exact sample size satisfies $\log n_r \asymp k \log(ed/k)$, as claimed. □

21 Scope

The theorem concerns the globally conditioned neural posterior itself, including arbitrary mixtures among tied variational minimizers. It is not a statement about posterior means or kernel ridge

prediction. The natural exponent is proved here with slowly increasing depth. For any fixed depth, the same natural-scale lower bound remains open. The lower polynomial bound on k ensures that the relevant Fourier levels exceed the uniform coefficient threshold $\exp(C/\beta)$ and that the logarithmic active faces fit within the parity support.

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